ADFP Cell-Based IC Physical Design and Verification with Innovus

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2026.10.07

Outline

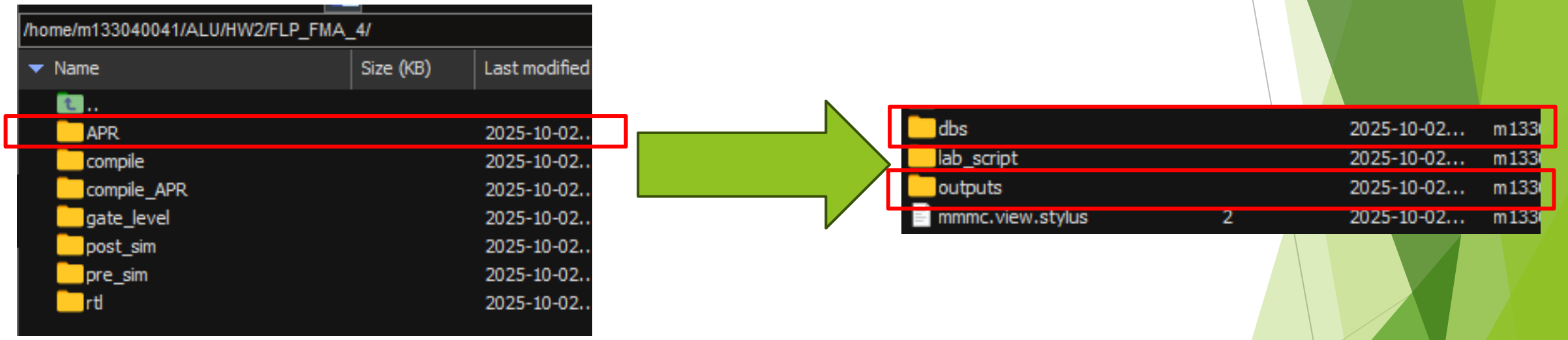
1. Pre-requirement
2. Assign Problem and Change Naming Rule Script
3. Design Import
4. Initial Step
5. Floor Plan
6. Power Plan
7. Power Route
8. Placement
9. CTS
10. Route
11. Post processing
12. Post Layout Simulation
13. HomeWork Requirment

Pre-requirement

- Gate-Level netlist (Verilog)
- TestBench and Pattern SDC
- constraints
- Physical Library (ADFP製程LEF)
- Timing Library (ADFP製程LIB)
- Multi Mode Multi Corner MMMC.view.stylus

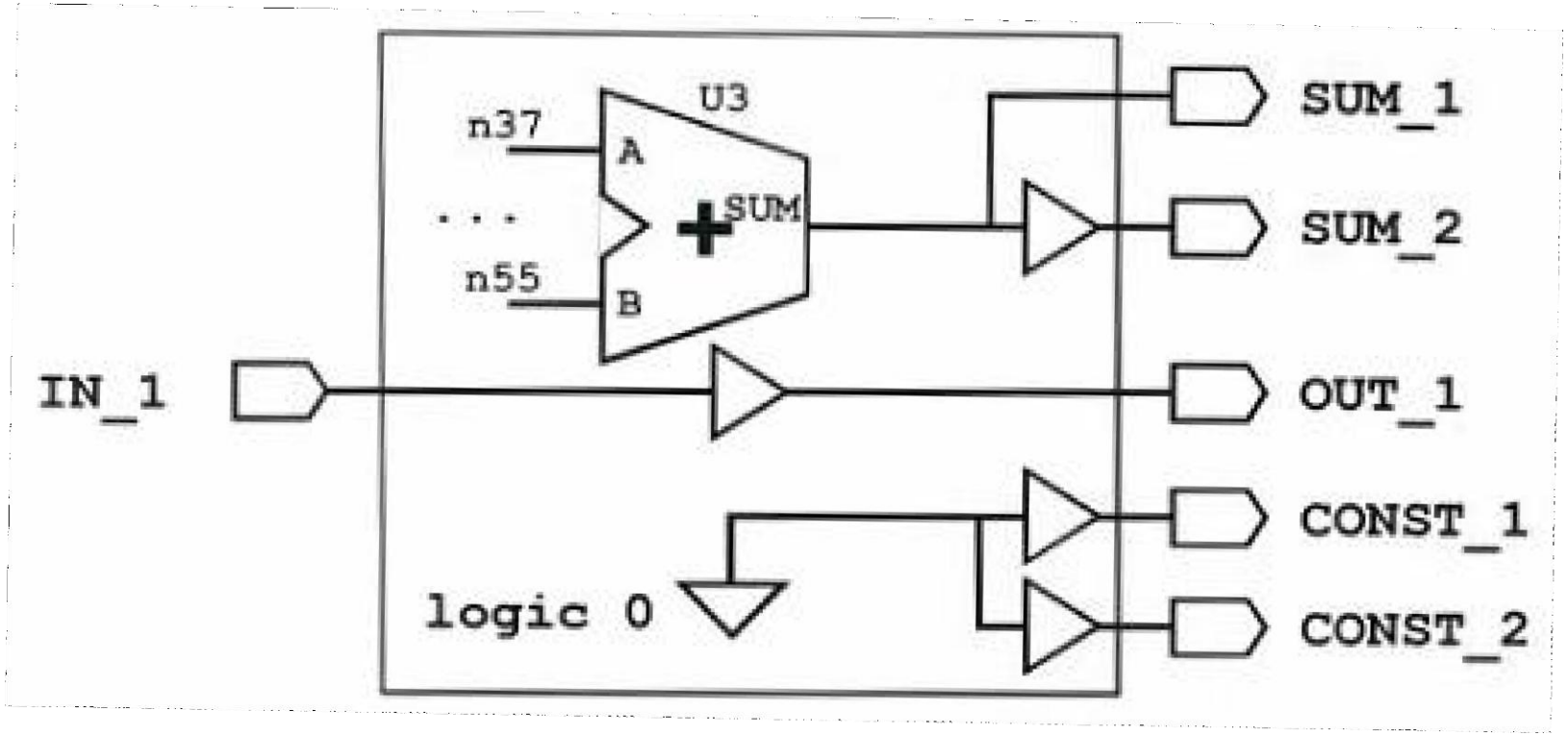
資料夾建立

- 1.pipeline版本的資料夾底下建立一個APR的資料夾
- 2.APR資料夾內建立dbs(存檔存放位置)outputs資料夾，同時把script和mmmc放入資料夾內



Assign Problem and Change Naming Rule Script(1/2)

- The syntax of “assign” may cause problems in the LVS



- To ensure that your final netlist does not contain assign statements,separate the multiple port nets
- during compile. (一定要在合成前做以下指令在進行合成)
- dc_shell > set_fix_multiple_port_nets –all –buffer_constants [get_designs *]

Assign Problem and Change Naming Rule Script(2/2)

```
assign \A[19] = A[19];
assign \A[18] = A[18];
assign \A[17] = A[17];
assign \A[16] = A[16];
assign \A[15] = A[15];
assign ABSVAL[19] = \A[19];
assign ABSVAL[18] = \A[18];
assign ABSVAL[17] = \A[17];
assign ABSVAL[16] = \A[16];
assign ABSVAL[15] = \A[15];
```



```
BUFX1 X37X( .I(A[19]), .Z(ABSVAL[19]) );
BUFX1 X38X( .I(A[18]), .Z(ABSVAL[18]) );
BUFX1 X39X( .I(A[17]), .Z(ABSVAL[17]) );
BUFX1 X40X( .I(A[16]), .Z(ABSVAL[16]) );
BUFX1 X41X( .I(A[15]), .Z(ABSVAL[15]) );
```

- The wrong naming rules may cause problems in the LVS
- 在合成完後加入以下指令
- set bus_inference_style {%s[%d]}
- set bus_naming_style {%s[%d]}
- set hdlout_internal_busses true
- change_names -hierarchy -rule verilog
- define_name_rules name_rule -allowed "A-Z a-z 0-9 _" -max_length 255 -type cell
- define_name_rules name_rule -allowed "A-Z a-z 0-9 _[]" -max_length 255 -type net
- define_name_rules name_rule -map {{"*cell*""cell"}}
- define_name_rules name_rule -case_insensitive
- change_names -hierarchy -rules name_rule

```
Warning: Verilog 'assign' or 'tran' statements are written out. (VO-4)
```

SDC File

需要將合成產生的.sdc檔案內的這幾行註解

```
set_units -time ns -resistance kOhm -capacitance pF -voltage V -current mA
set_operating_conditions -max ss0p72vm40c -max_library                    \
N16ADFP_StdCellss0p72vm40c_ccs\
                         -min ff0p88v125c -min_library                    \
N16ADFP_StdCellff0p88v125c_ccs
set_wire_load_mode top
set_wire_load_model -name ZeroWireload -library N16ADFP_StdCellss0p72vm40c_ccs
set_max_area 0
set_ideal_network [get_ports clk]
```

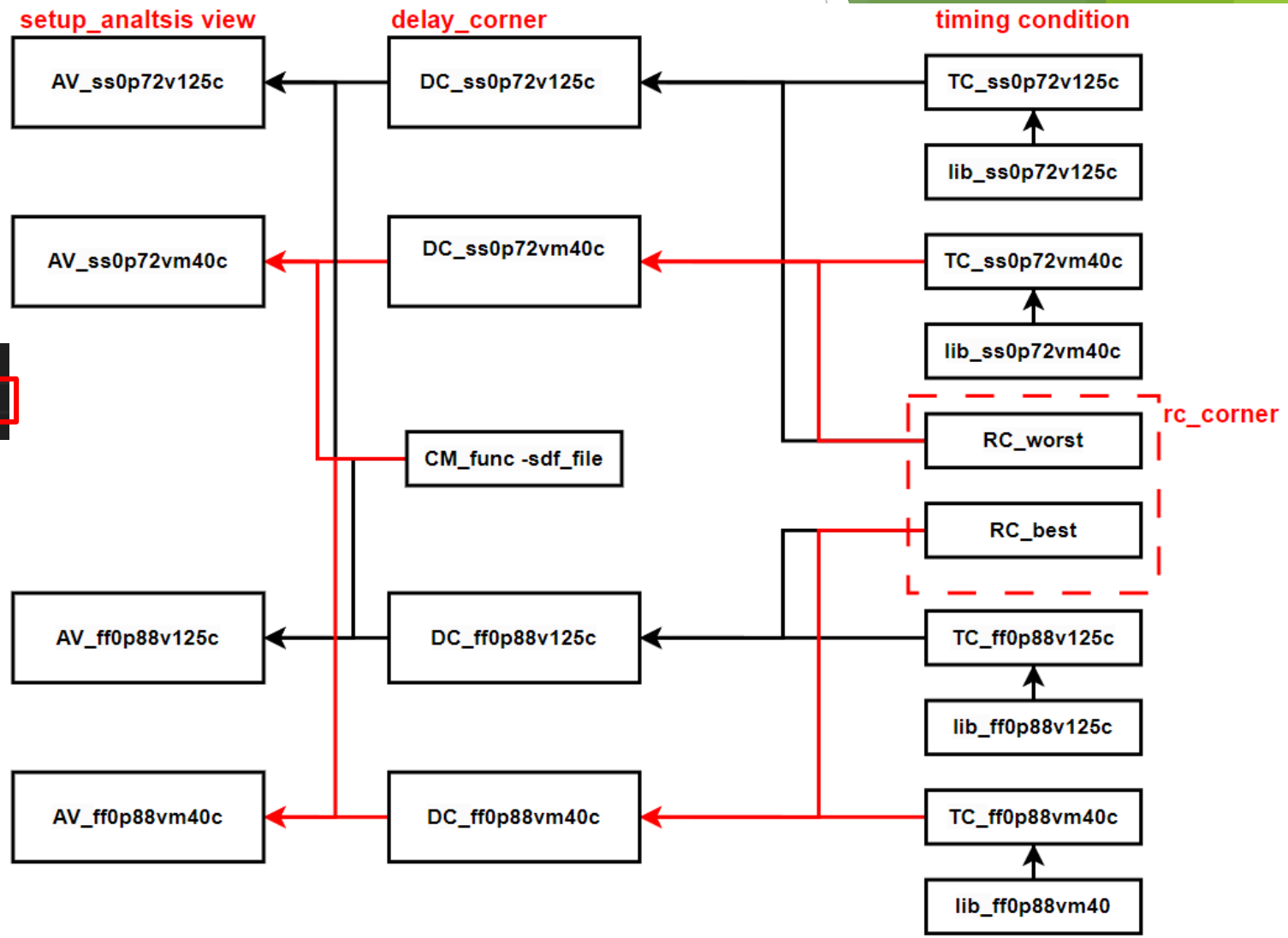


```
1  ##############################################################################
2
3  # Created by write_sdc on Wed Sep 11 16:12:33 2024
4
5  ##############################################################################
6  set sdc_version 2.1
7
8  #set_units -time ns -resistance kOhm -capacitance pF -voltage V -current mA
9  #set_operating_conditions -max ss0p72vm40c -max_library                    \
10 #N16ADFP_StdCellss0p72vm40c_ccs\
11 #                         -min ff0p88v125c -min_library                    \
12 #N16ADFP_StdCellff0p88v125c_ccs
13 #set_wire_load_mode top
14 #set_wire_load_model -name ZeroWireload -library N16ADFP_StdCellss0p72vm40c_ccs
15 #set_ideal_network [get_ports clk]
16 create_clock [get_ports clk]  -period 2.5  -waveform {0 1.25}
17 set_clock_latency 0.2  [get_clocks clk]
18 set_clock_uncertainty 0.02  [get_clocks clk]
19 set_clock_transition -max -rise 0.1 [get_clocks clk]
20 set_clock_transition -max -fall 0.1 [get_clocks clk]
21 set_clock_transition -min -rise 0.1 [get_clocks clk]
22 set_clock_transition -min -fall 0.1 [get_clocks clk]
```

Multi Mode Multi Corner MMMC.view.stylus

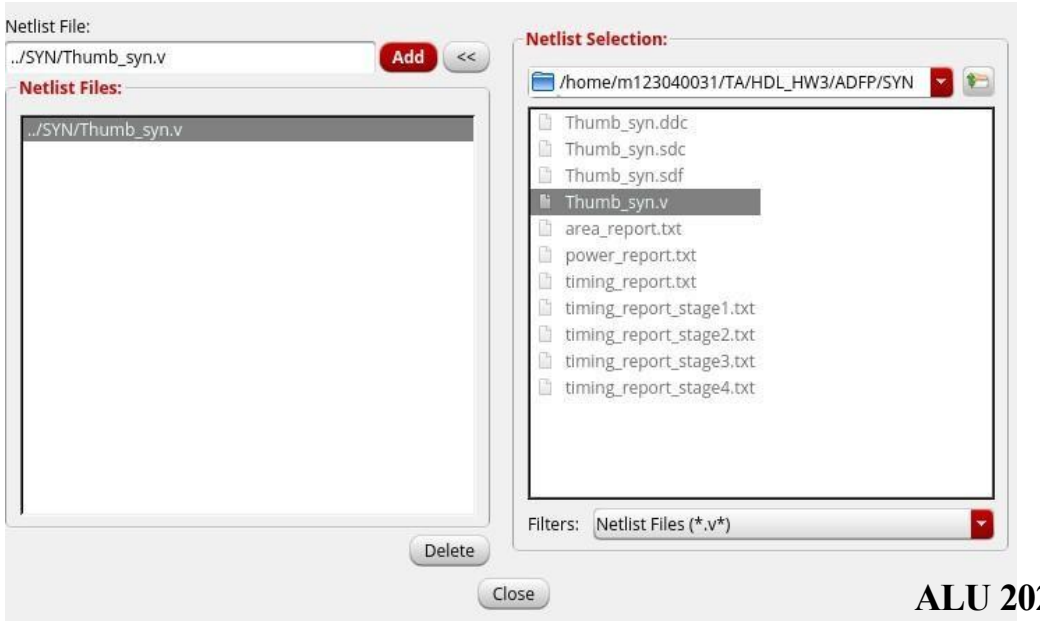
sdc_file(XXX_syn.sdc)

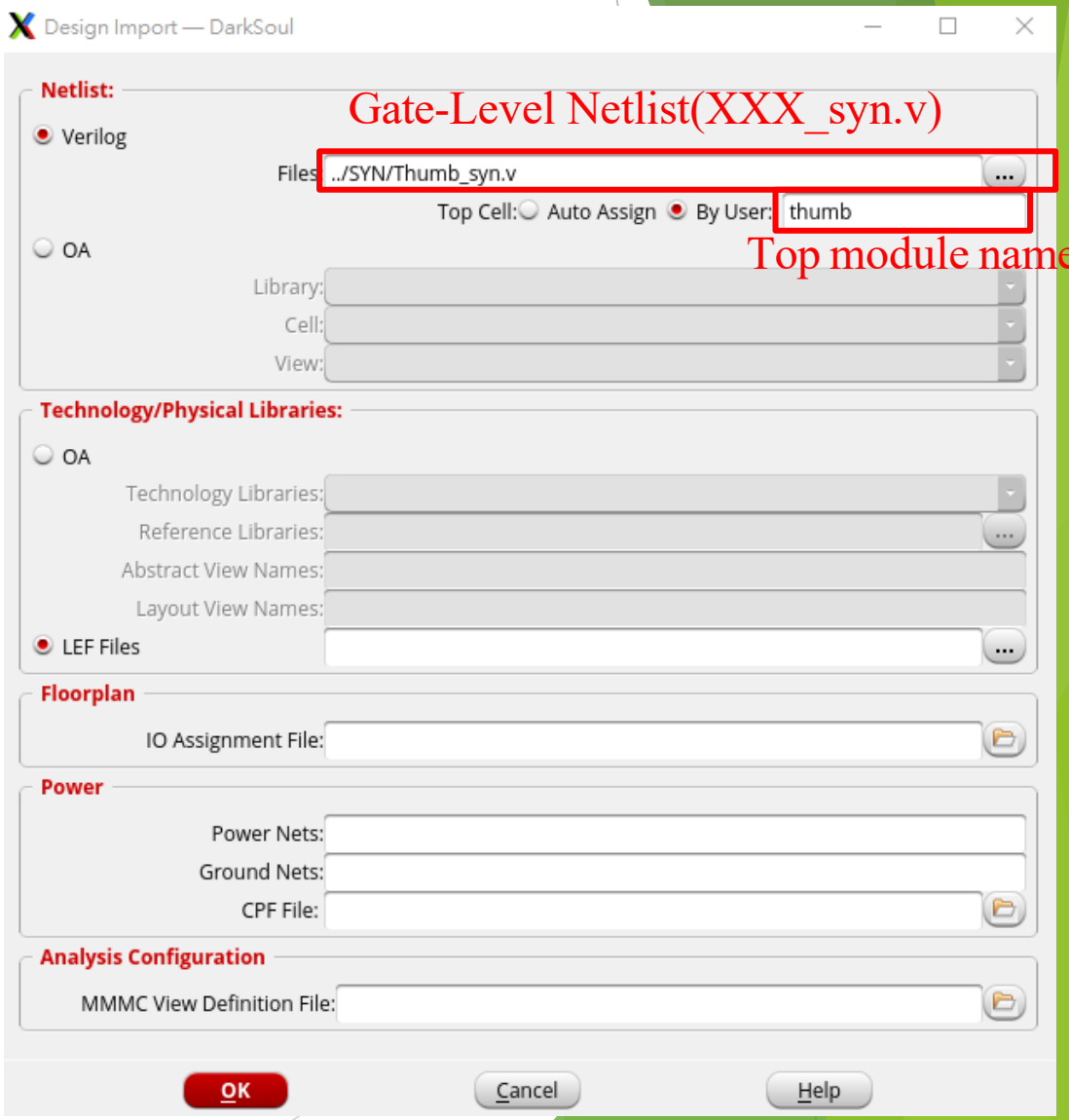
```
create_constraint_mode -name CM_func\
    -sdc_files /home/m123040031/TA/HDL_HW3/ADFP/SYN/Thumb_syn.sdc
```



Design import(1/4)

- cd 進入APR資料夾，輸入tcsh
- Terminal 下輸入innovus -stylus
- 1.File → Import Design → Netlist → Verilog → Files → … →select your file and click Add
- 2.Select By User,fill your_design_name





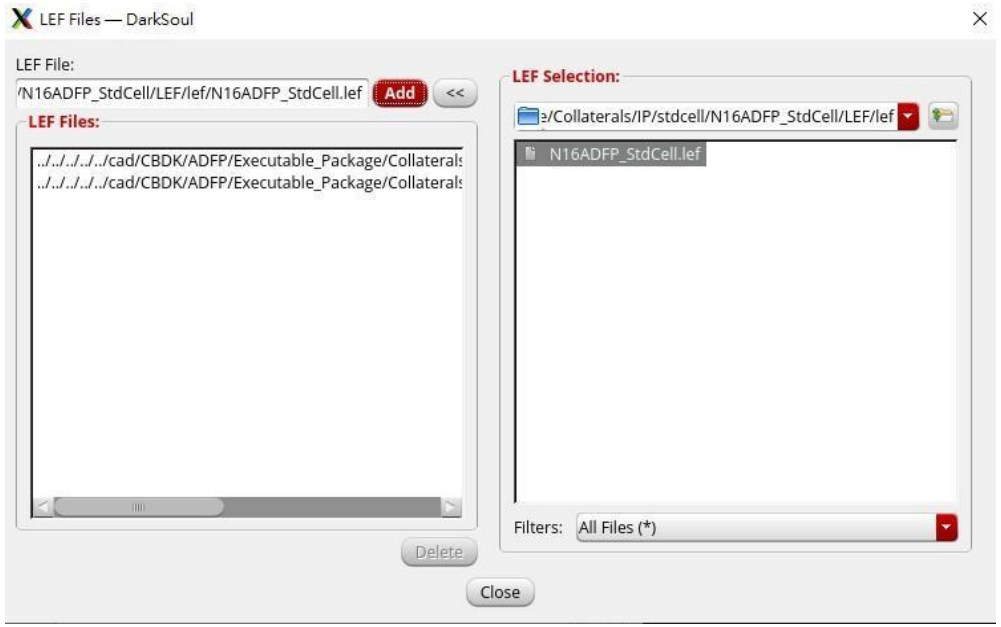
Design import(2/4)

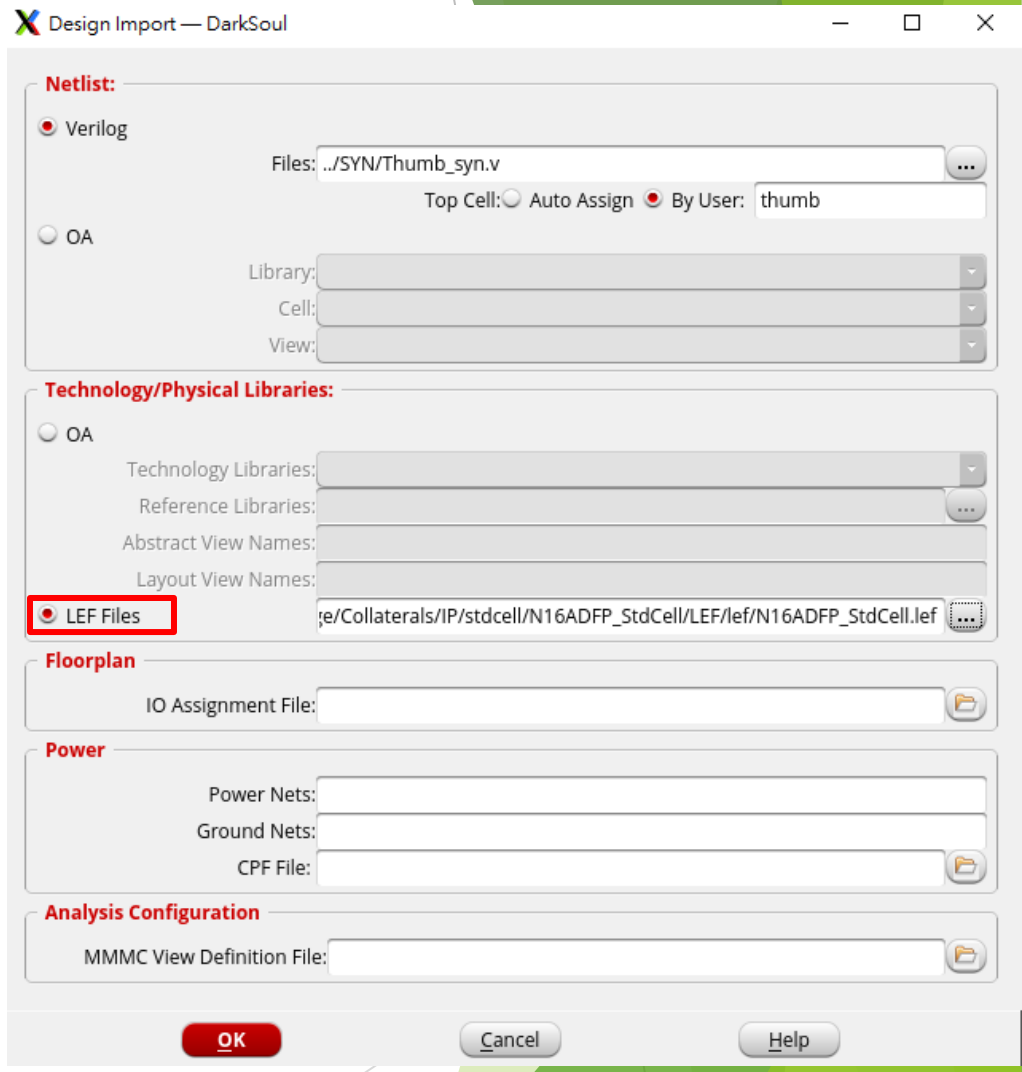
▶ 3. Load process file

▶LEF Files → … → Filters：All Files(*) → add lef files

註：製程的technology lef一定要放第一個，其餘的無順序之分，不同製程有不同的名字，如果不知道的話就一個一個試，錯誤的話最後設定

完成後就會出現錯誤。(此範例為N16ADFP_APR_Innovus_11M.10a.tlef)

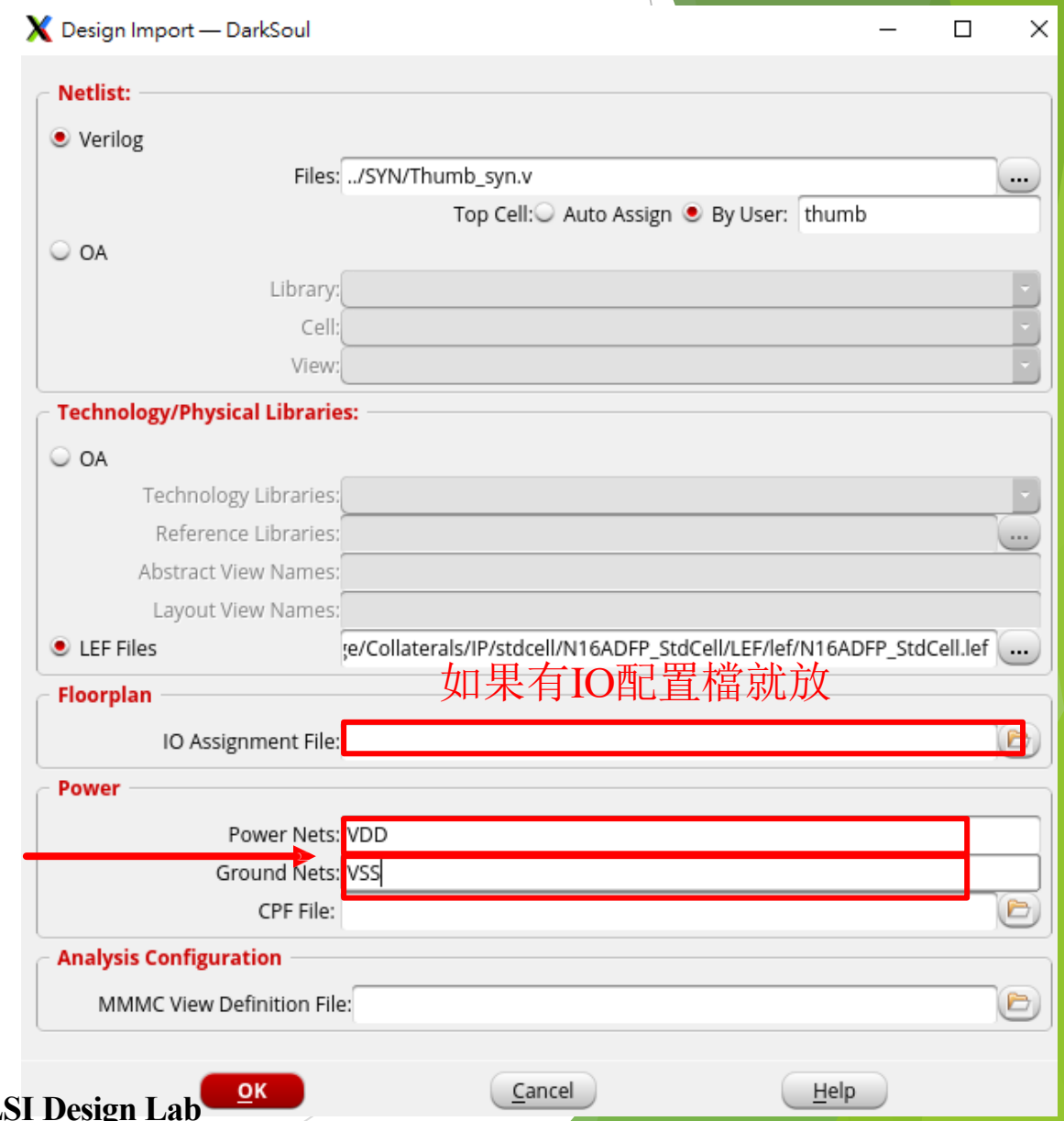




/cad/CBDK/ADFP/Executable_Package/Collaterals/Tech/APR/N16ADFP_APR_Innovus/N16ADFP_APR_Innovus_11M.10a.tlef
/cad/CBDK/ADFP/Executable_Package/Collaterals/IP/stdcell/N16ADFP_StdCell/LEF/lef/N16ADFP_StdCell.lef

Design import(3/4)

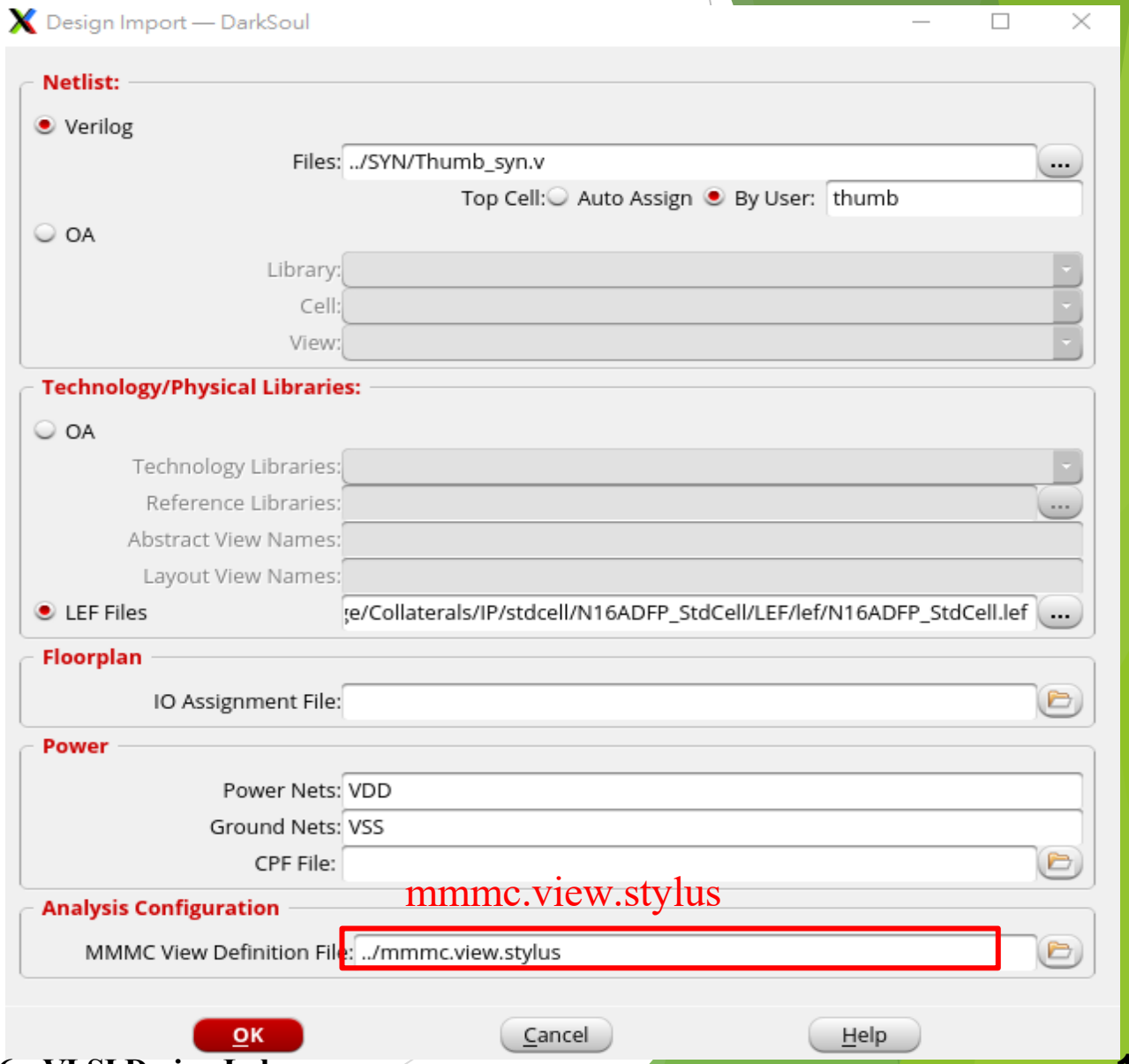
- 4.(Optional) IO Assignment File → add IO file
- 5.Power → Power Nets:VDD;Ground Nets:VSS



TSMC： VDD VSS
UMC： VCC GND

Design import(4/4)

- 6. mmmc.view.stylus



Design import

- 如果import design 後有出現error message，1.先看是什麼地方出錯，並修正
- 2.重開innovus再 import一次

Initial Step(1/3)

- 1.Connect global power
- Power → Connect Global Nets → Connect:Pin → Pin Name(s): VDD;

  Scope:Apply All; To Global Net: VDD → Add to List → Do it again but change VDD to VSS
- Power → Connect Global Nets → Connect:Pin → Pin Name(s): VPP;

  Scope:Apply All; To Global Net: VDD → Add to List
- Power → Connect Global Nets → Connect:Pin → Pin Name(s): VBB;

  Scope:Apply All; To Global Net: VSS → Add to List
- Power → Connect Global Nets → Connect:Tie High → Scope:Apply All;

  To Global Net: VDD → Add to List
- Power → Connect Global Nets → Connect:Tie Low → Scope: All;

  To Global Net: VSS → Add to List→ Apply → Check → CloseApply

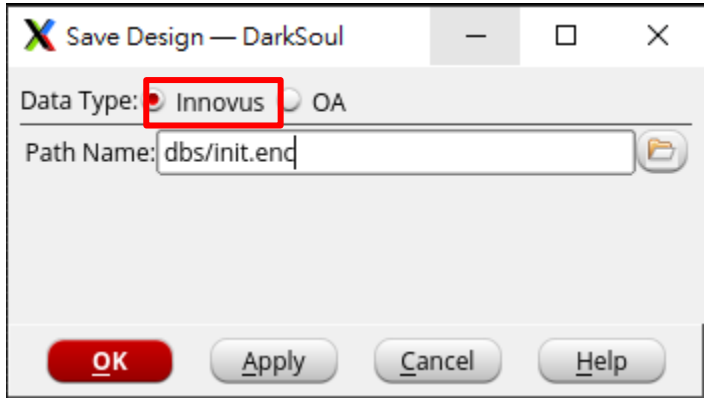
  (Tie high Tie Low不要打Pin name)
- Apply
- Cancel



Initial Step(2/3)

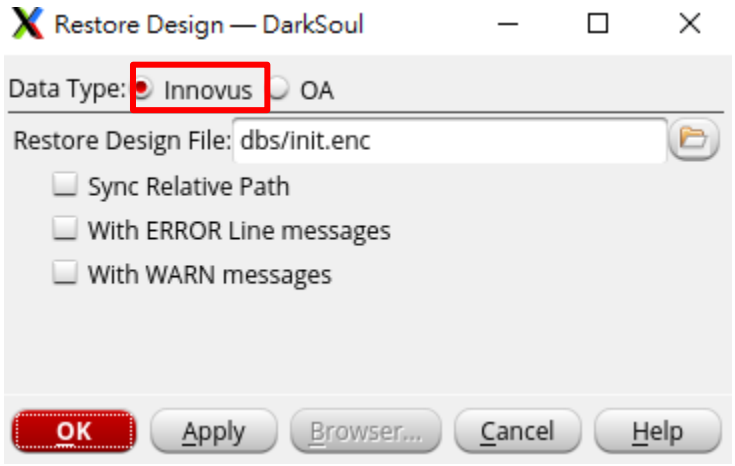
- innovus > source -quiet lab_script/config.tcl
- innovus > source lab_script/config_cts.tcl
- Save Design

  File →Save Design



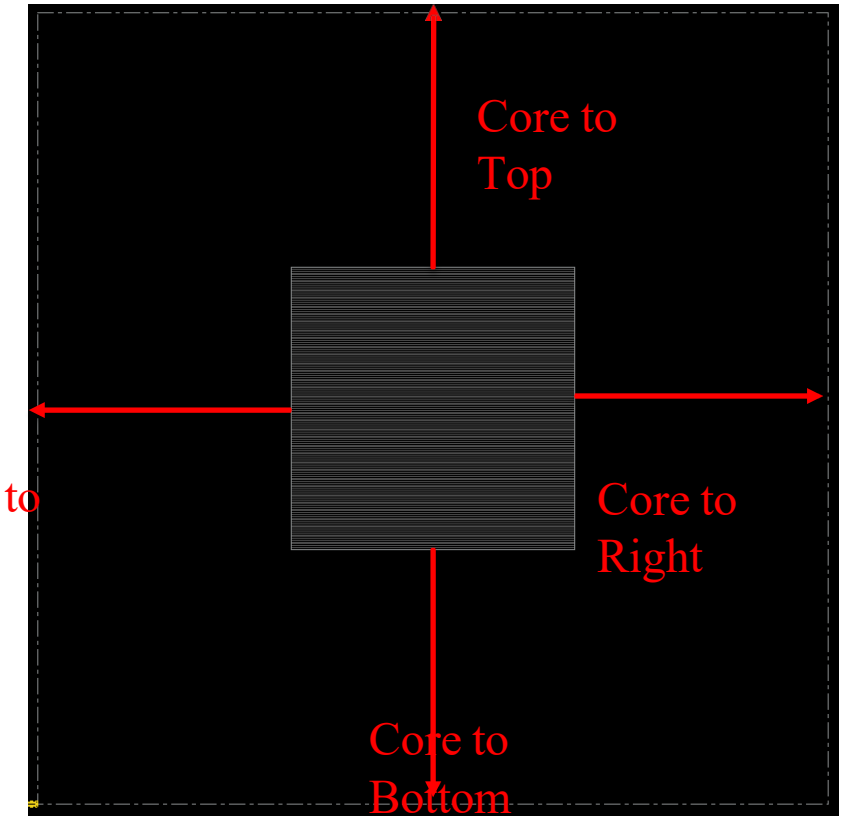
- Restore Design

  File →Restore Design



Initial Step(3/3)

- 3.Decide the overview of your design.
- Floorplan → Specify Floorplan… → …
- Ratio (H/W):1 ; Core Utilization:0.6 ; (建議0.6-0.8，因為這次的cell area較小)
- Core to Left:80; Core to Right:80; Core to Top:80; Core to Bottom:80;

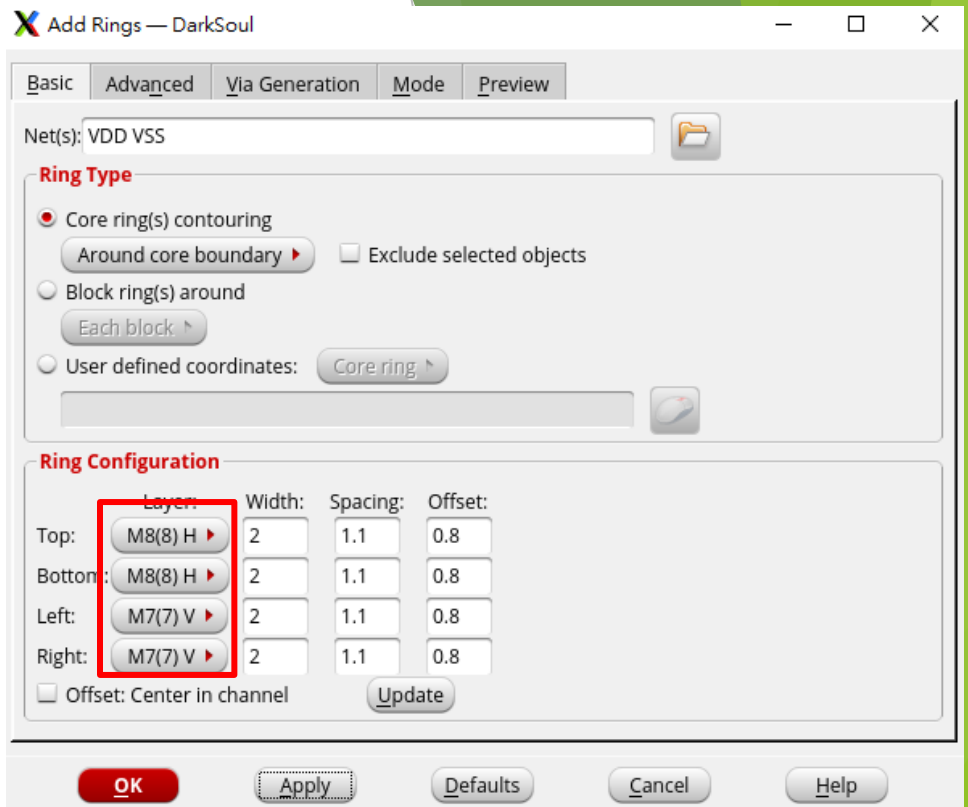


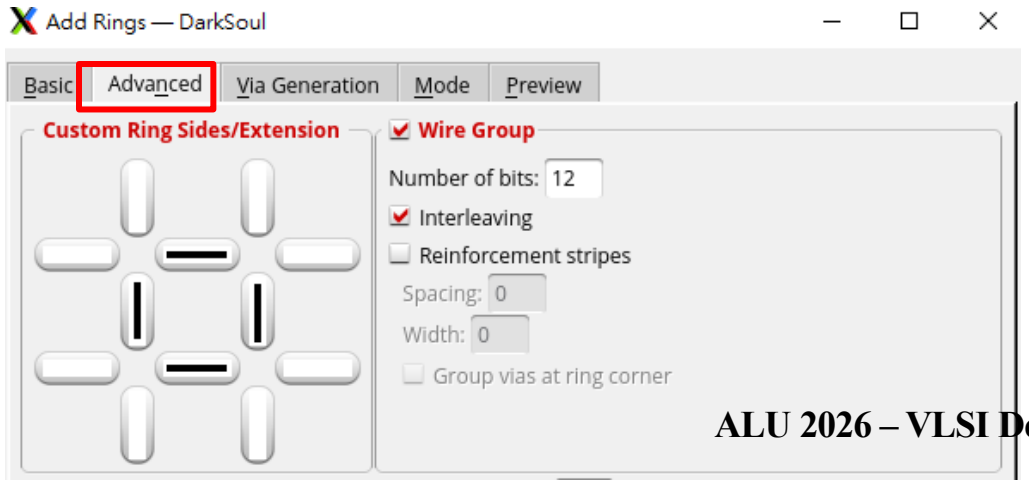


註：如果有PAD的話要調整適當的距離，如果Core跟PAD 太接近， 在後面接Power Ring及Route後，有可能會DRC錯誤

PowerPlan(1/9)

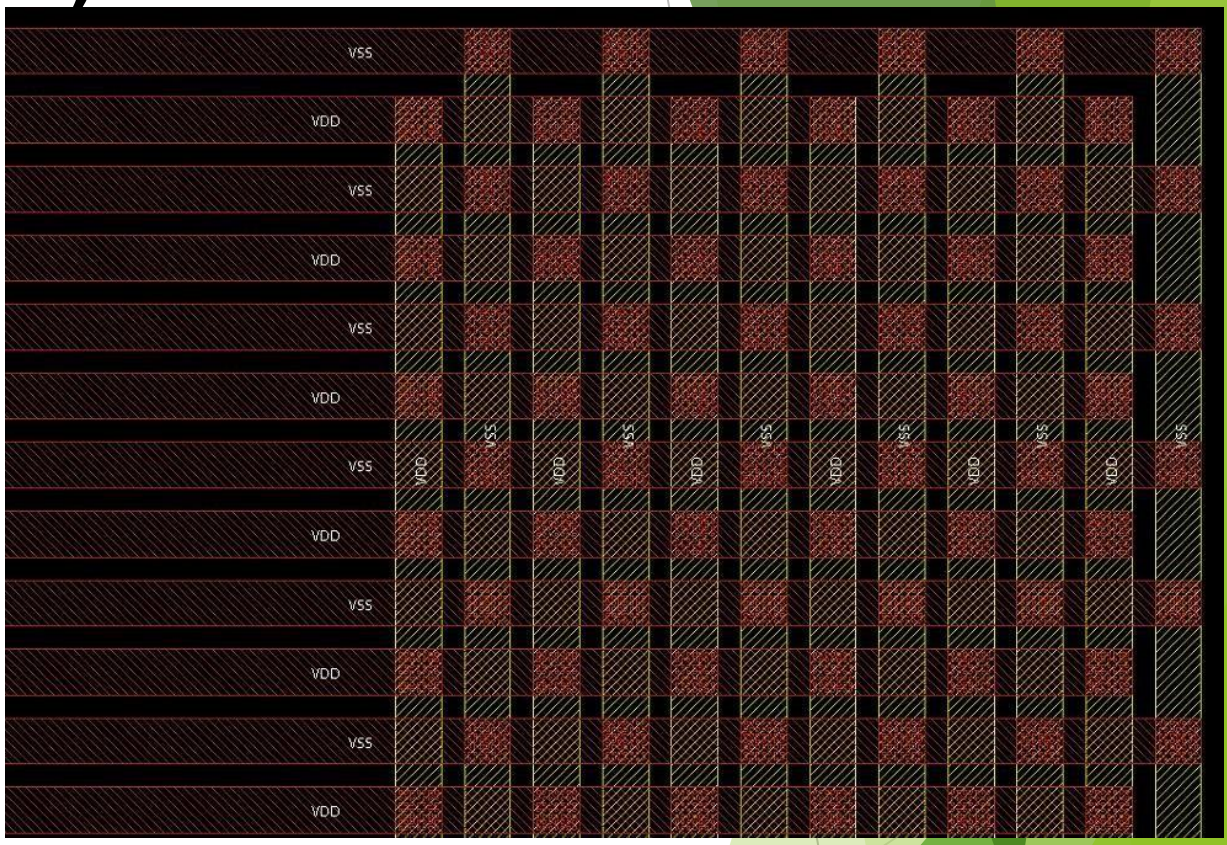
1. innovus > source lab_script/pns.tcl
2. Create Power Ring ---- To give Power to chip
3. Power → Power Planning → Add Rings → Basic
   - Nets: VDD VSS
   - In Ring Configuration,change Top & Bottom to METAL8, Left & Right to METAL7;
   - Set Width to 2 and Spacing to 1.1
4. Change to Advanced
   - Select Wire Group,set Numbers of Bits to 12
   - Select Interleaving



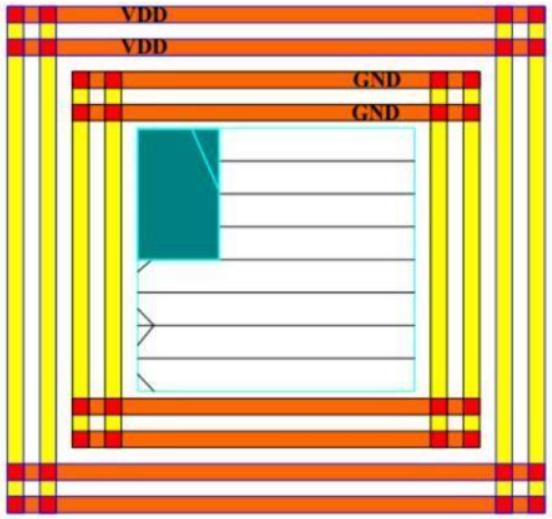


PowerPlan(2/9)

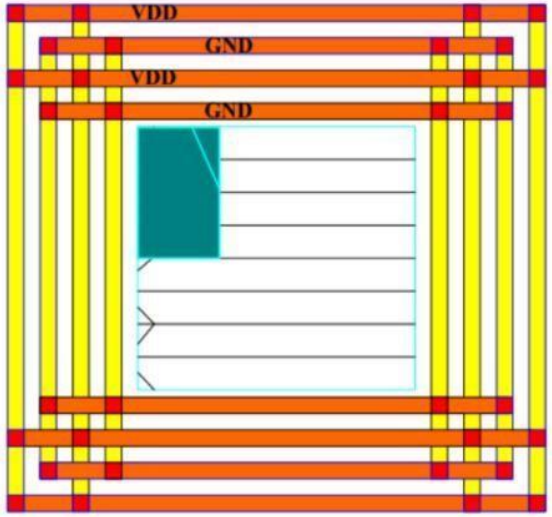
- What's Interleaving?



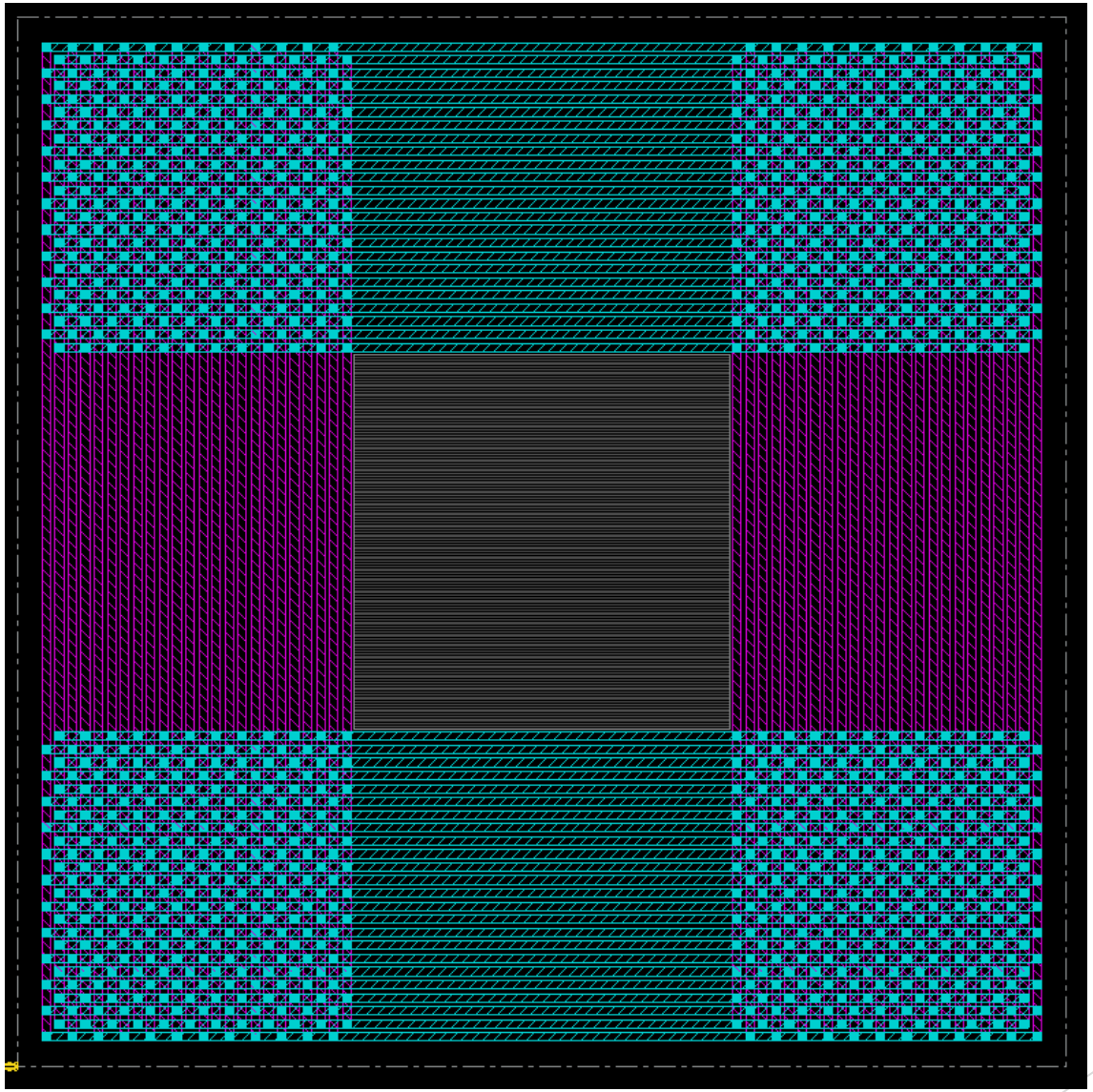
✓Use wire group
no interleaving
✓number of bits = 2



✓Use wire group
✓interleaving
✓number of bits = 2



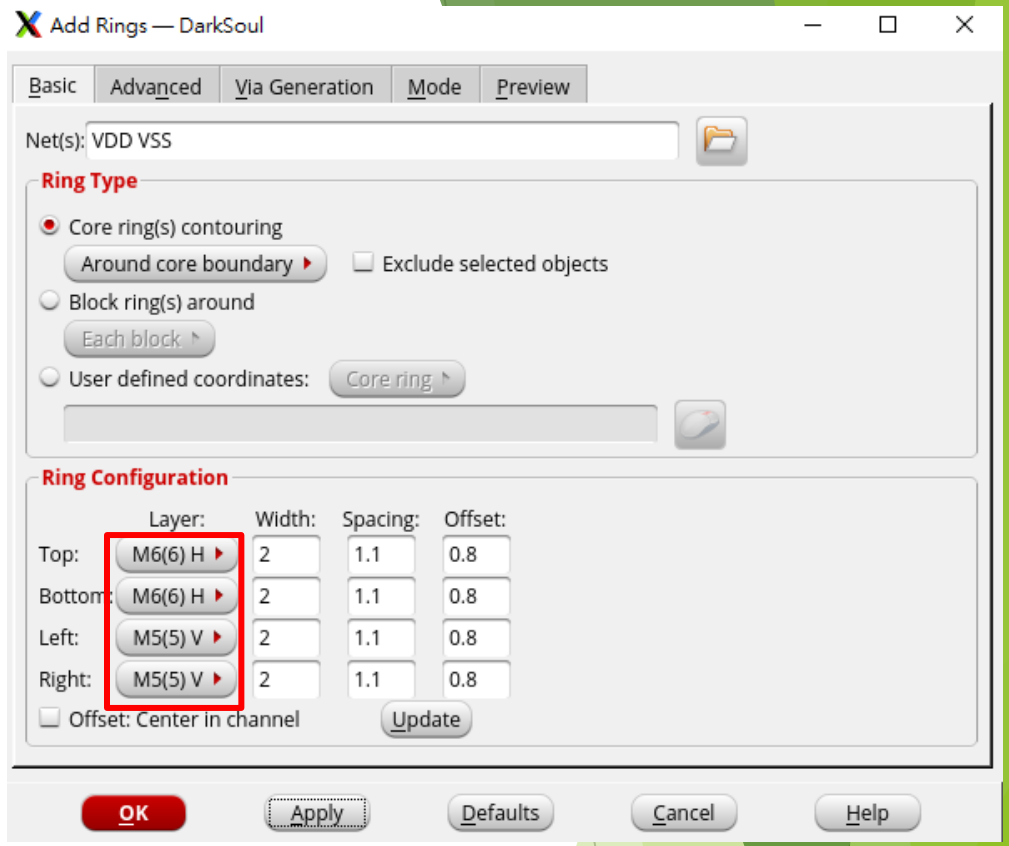
PowerPlan(3/9)

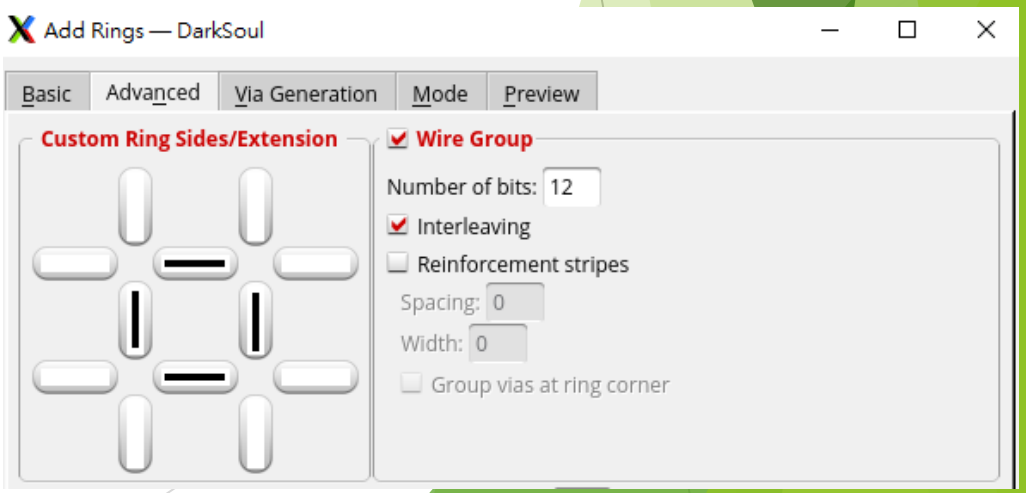


PowerPlan(4/9)

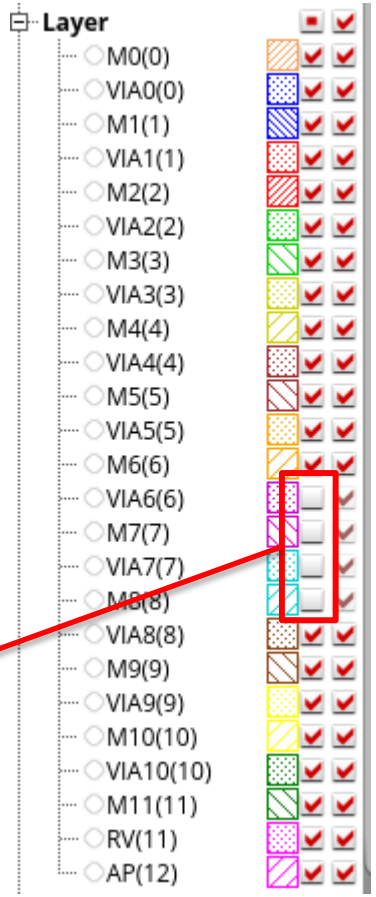
- 5. Create Power Ring ---- To give Power to chip
- 6. Power → Power Planning → Add Rings → Basic
  - Nets: VDD VSS
  - In Ring Configuration,change Top & Bottom to METAL6, Left & Right to METAL5;
  - Set Width to 2 and Spacing to 1.1
- 5.Change to Advanced
  - Select Wire Group,set Numbers of Bits to 12
  - Select Interleaving
- 6. innovus > edit_trim_routes -nets {VDD VSS} -layer {M8 M7 M6 M5}

會把Power Ring 多餘的部分切掉

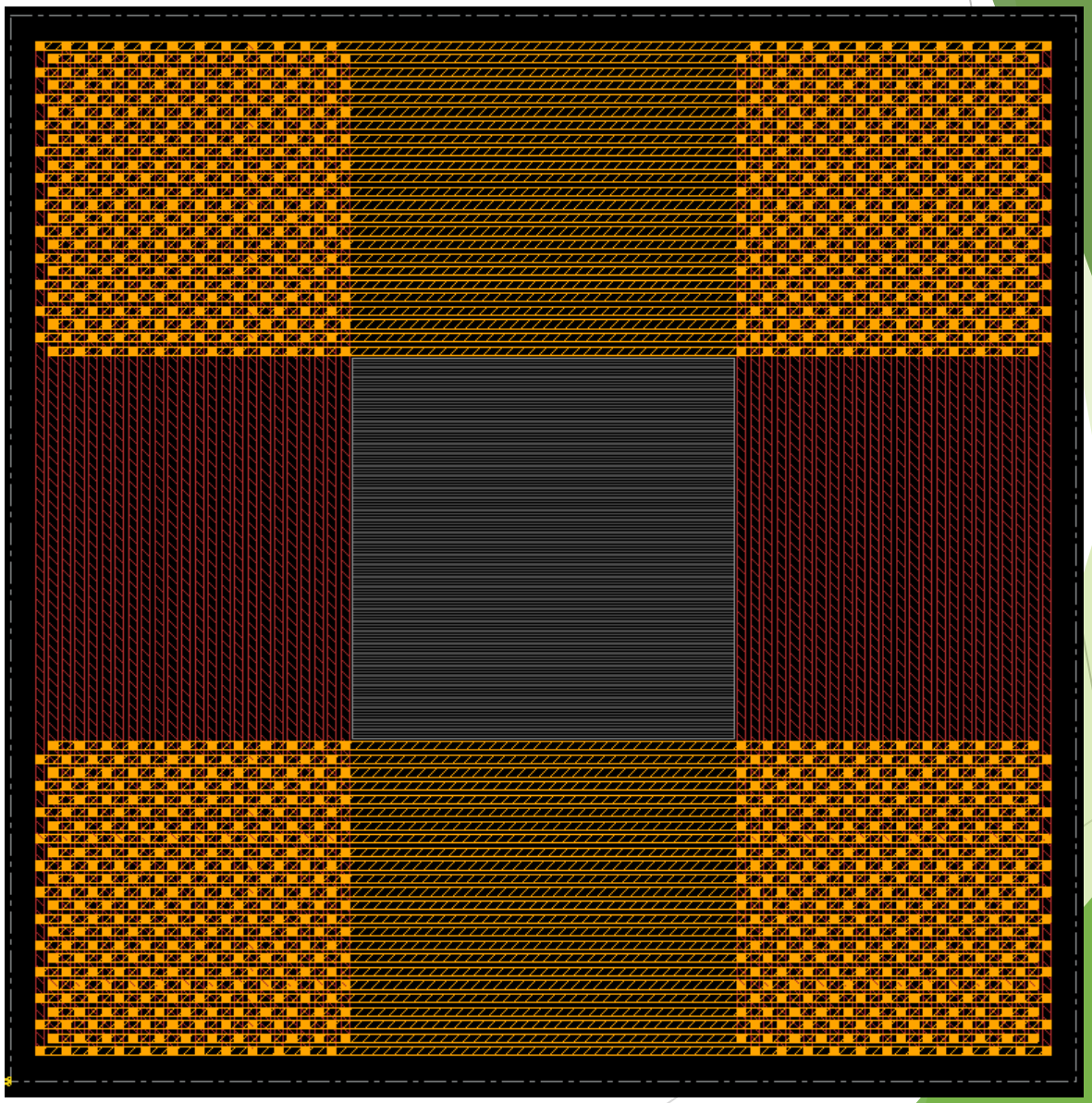




PowerPlan(5/9)

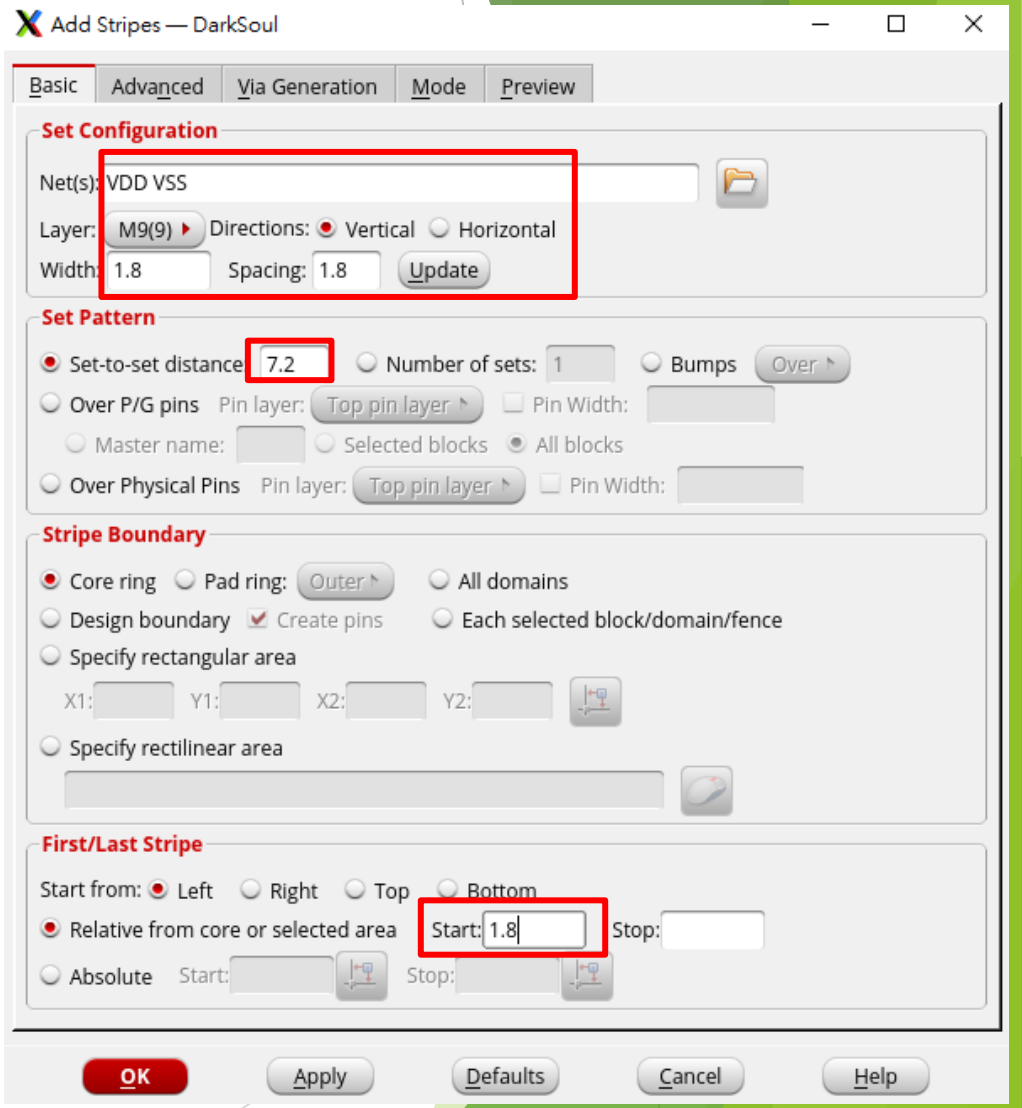


把這4層勾選掉才能看到M5, M6



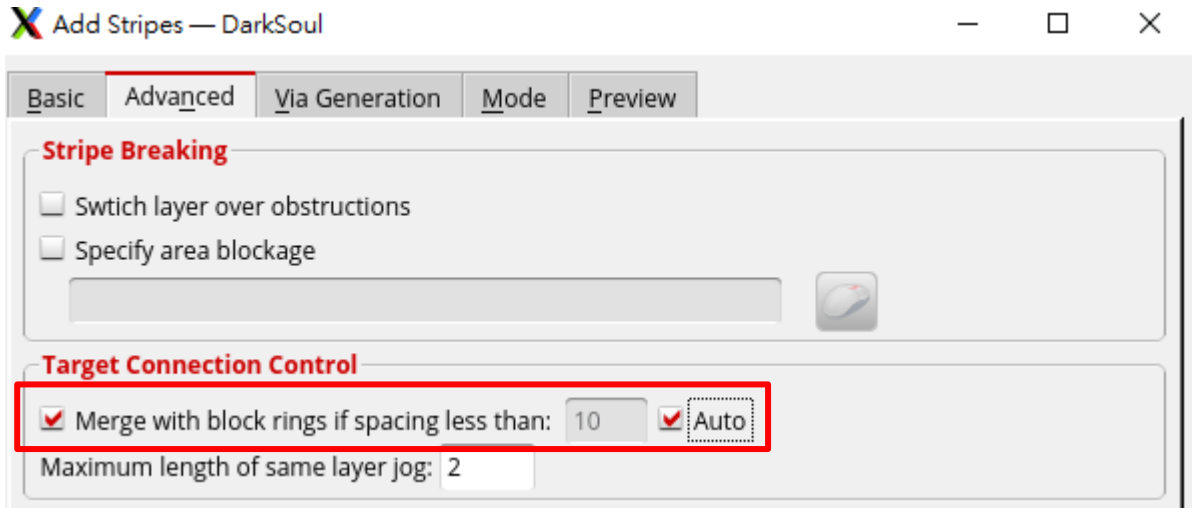
PowerPlan(6/9)

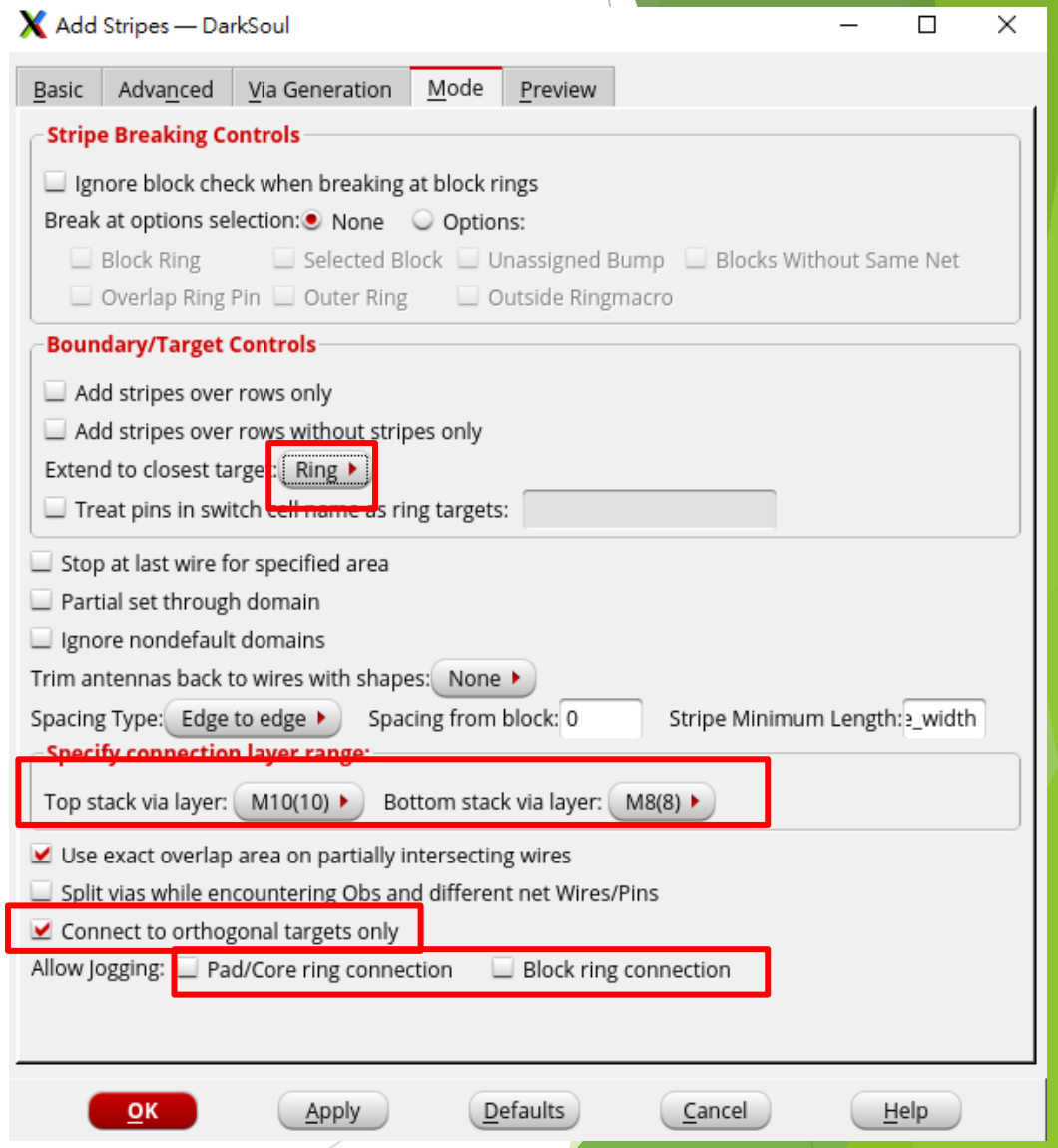
- 8. Add stripe
  - The deeper part of your design may not have enough power from power ring, so you need to add stripe to boost the voltage.
  - Vertical stripe is recommended.
- 9. Power → Power Planning → Add Stripe → Basic
  - Nets: VDD VSS
  - Layers: METAL9 and select Vertical;
  - Set Width to 1.8 and Spacing to 1.8
  - In Set Pattern, select Set-to-set distance and input 7.2
  - Select Relative from core or selected area
  - Start: 1.8
- 接續下一頁



PowerPlan(7/9)

10. Change to **Advanced** Page
    - Choose **Merge** with block ring if spacing less then : 10
    - Choose **Auto**
11. Change to **Mode**
    - Extend to closest target: **Ring**
    - Top Stack via layer: **METAL10** Bottom Stack via layer: **METAL8**
    - Choose **Connect to orthogonal** targets only
    - **un-choose** Pad/Core ring connection & Block ring connection



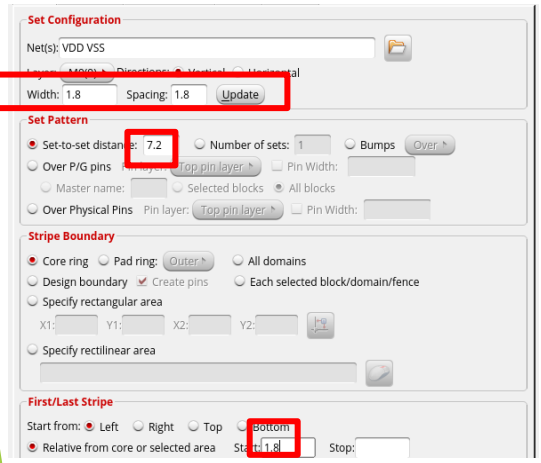


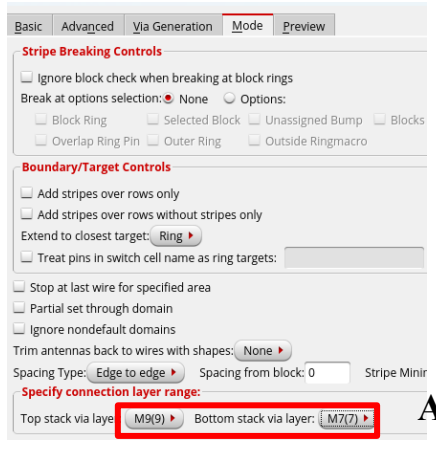
PowerPlan(8/9)

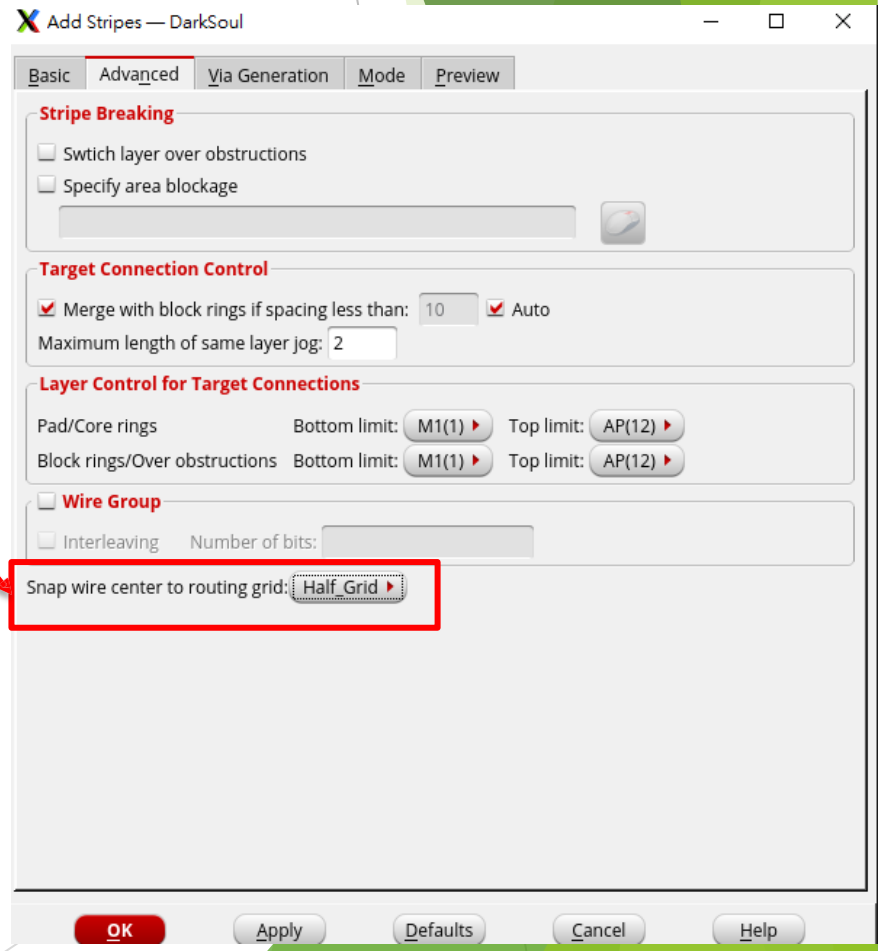
Basic Mode Advanced

Net	Metal layer	dir	width	space	set to set	start	Top stack via	Bottom stack via	Snap to Routing grid
VDD VSS	M9	V	1.8	1.8	7.2	1.8	M10	M8	
VDD	M8	H	0.864	0	2.88	0	M9	M7	Half_Grid
VSS	M8	H	0.864	0	2.88	1.44	M9	M7	Half_Grid
VDD	M7	V	0.24	0	7.2	7.2	M8	M6	Grid
VSS	M7	V	0.24	0	7.2	3.6	M8	M6	Grid
VDD	M6	H	0.24	0	7.2	0	M7	M5	Grid
VSS	M6	H	0.24	0	7.2	3.6	M7	M5	Grid
VDD	M5	V	0.24	0	7.2	0	M6	M4	Grid
VSS	M5	V	0.24	0	7.2	3.6	M6	M4	Grid

- 重複6~7的步驟做9次，並更改上面的內容
- 如果打完發現沒有長stripe，沒關係，如果想要有，把utilization設0.4再做一次





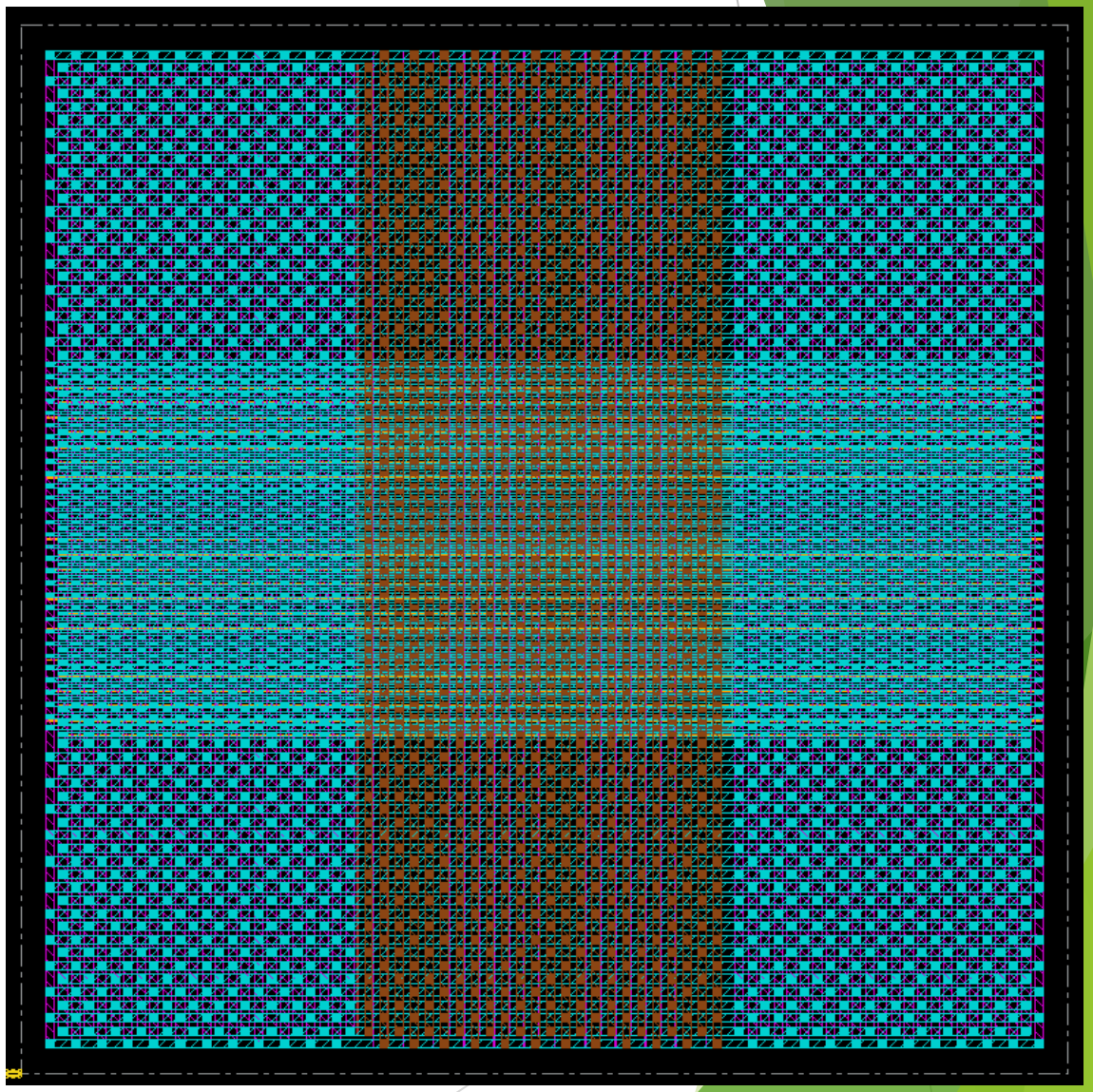


PowerPlan(9/9)

- 12. Check → CheckDRC… → OK

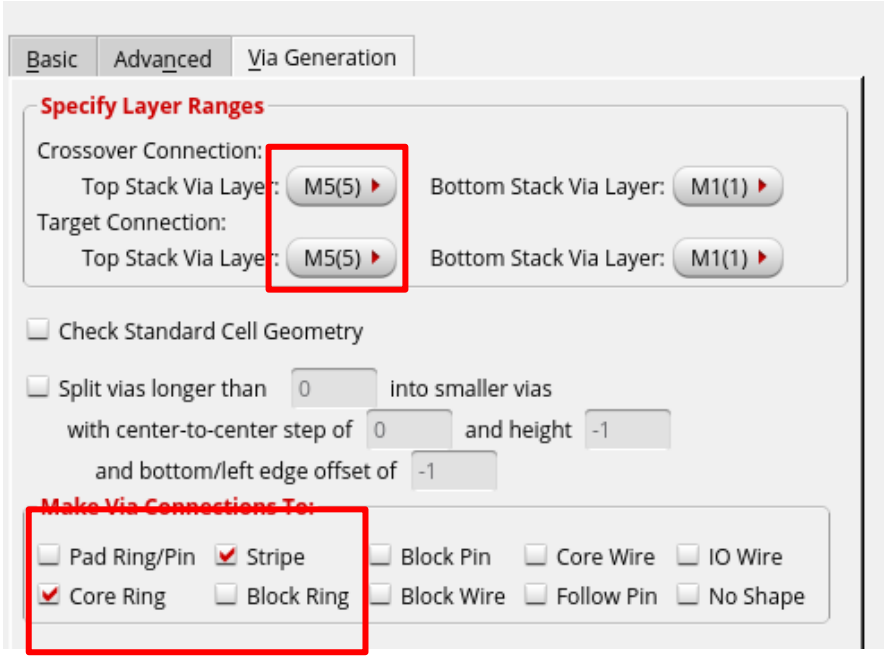
```
VERIFY DRC ..... Sub-Area: {194.432 222.208 222.208 248.736} 80 of 81
VERIFY DRC ..... Sub-Area : 80 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {222.208 222.208 248.940 248.736} 81 of 81
VERIFY DRC ..... Sub-Area : 81 complete 0 Viols.

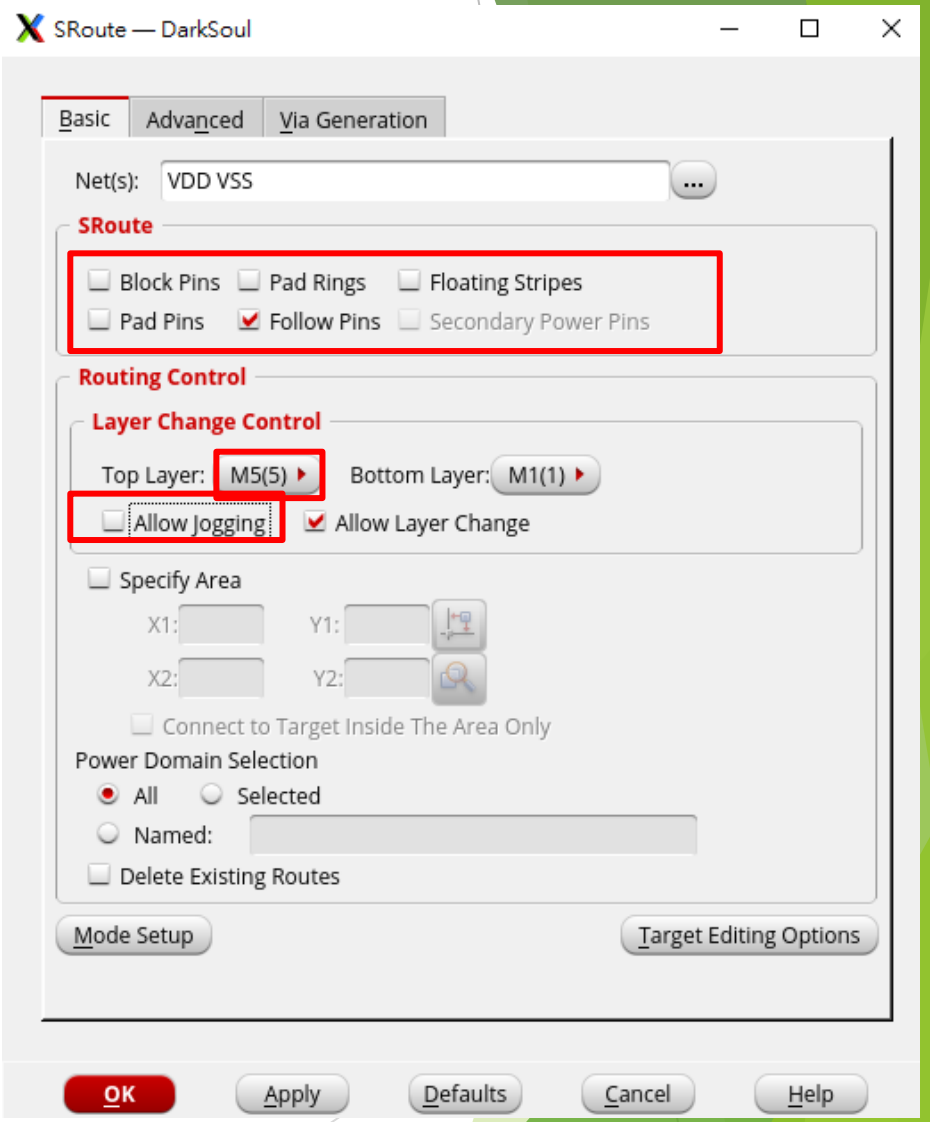
Verification Complete : 0 Viols.
```



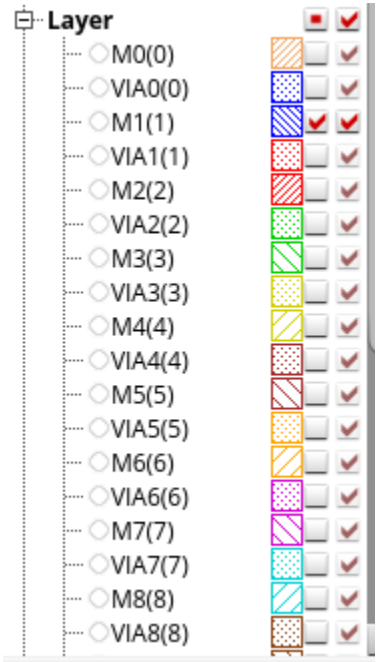
Power Route(1/8)

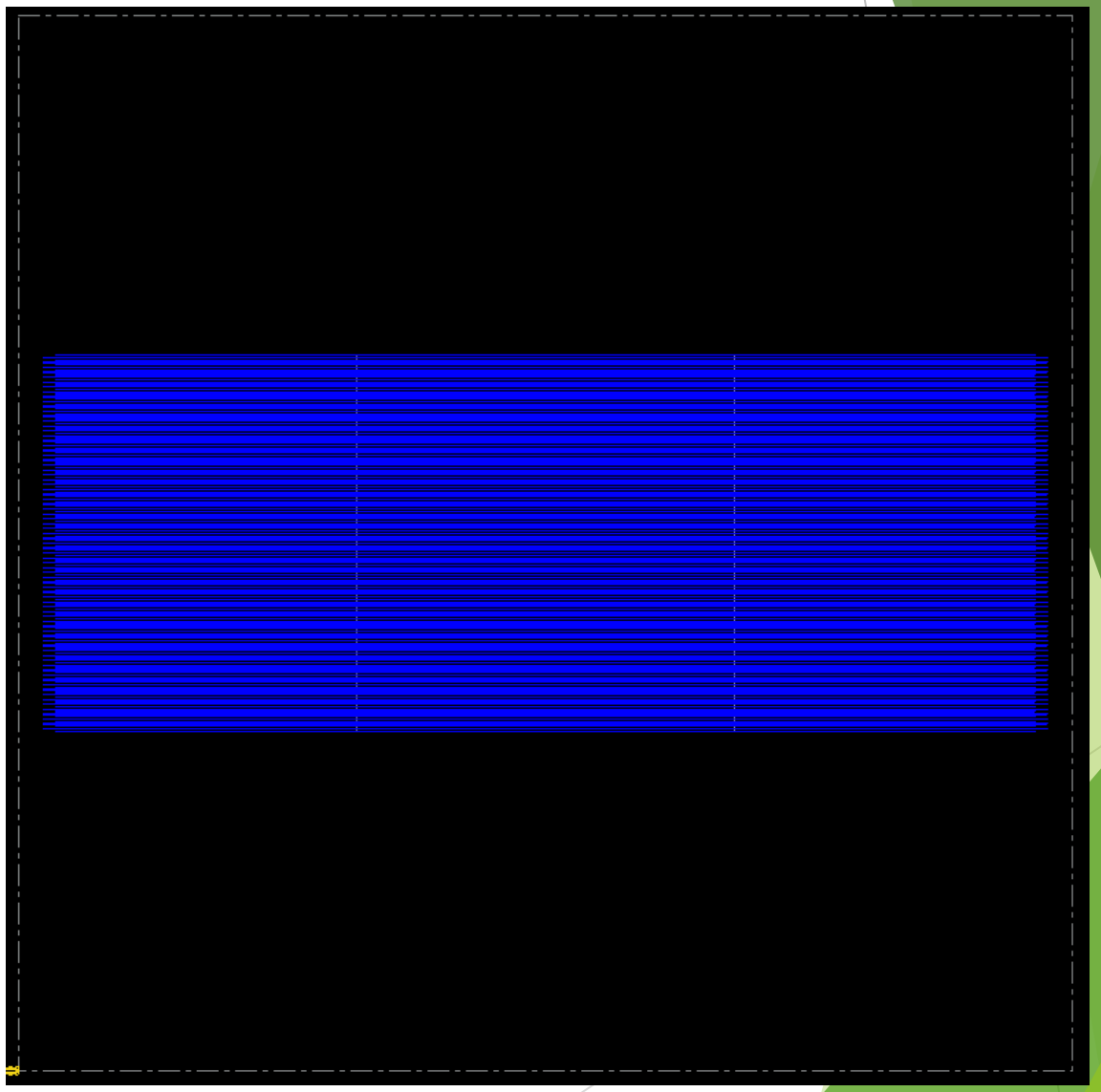
- What's power route? ---- Connect all cell to power ring
- 1. Route → Special Route → Basic
  - Net(s): VDD VSS
  - In SRoute, only select Follow pins
  - un-choose Allow jogging
- 2. Change to Via Generation
  - In Make Via Connection To, select Core Ring & Stripe
  - Crossover Connection: Top Stack Via Layer: METAL5





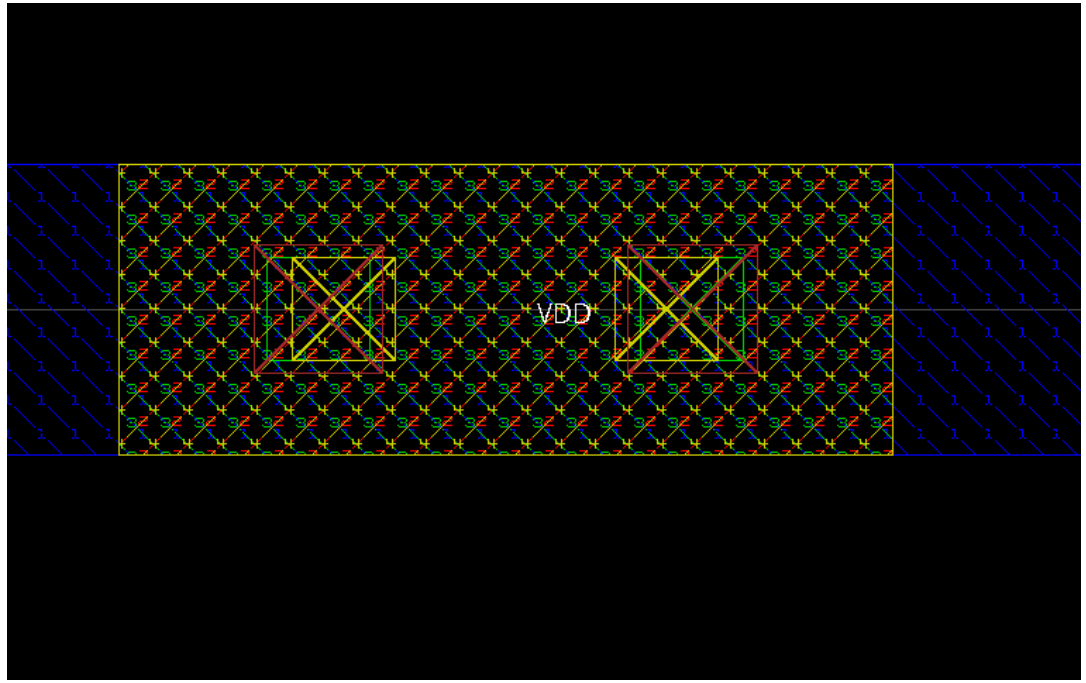
Power Route(2/8)

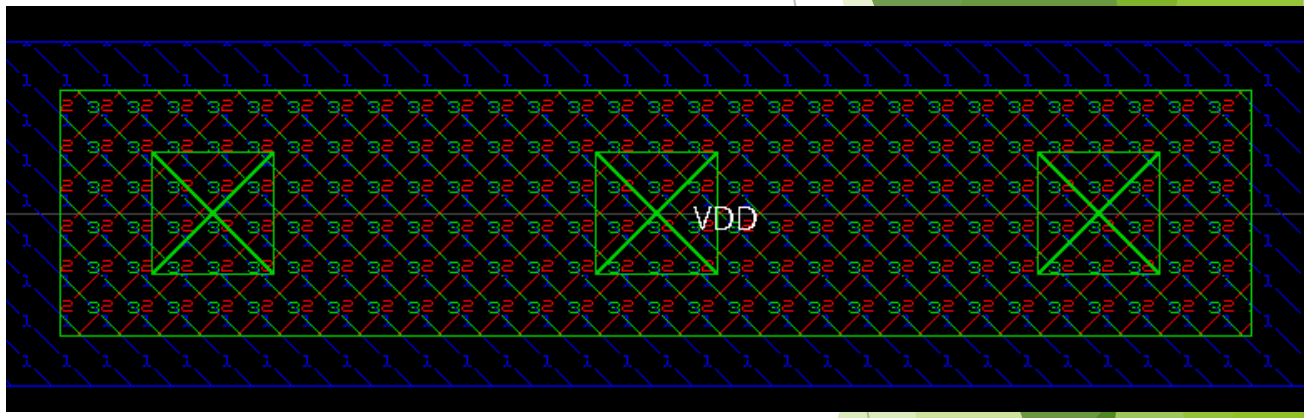




Power Route(3/8)

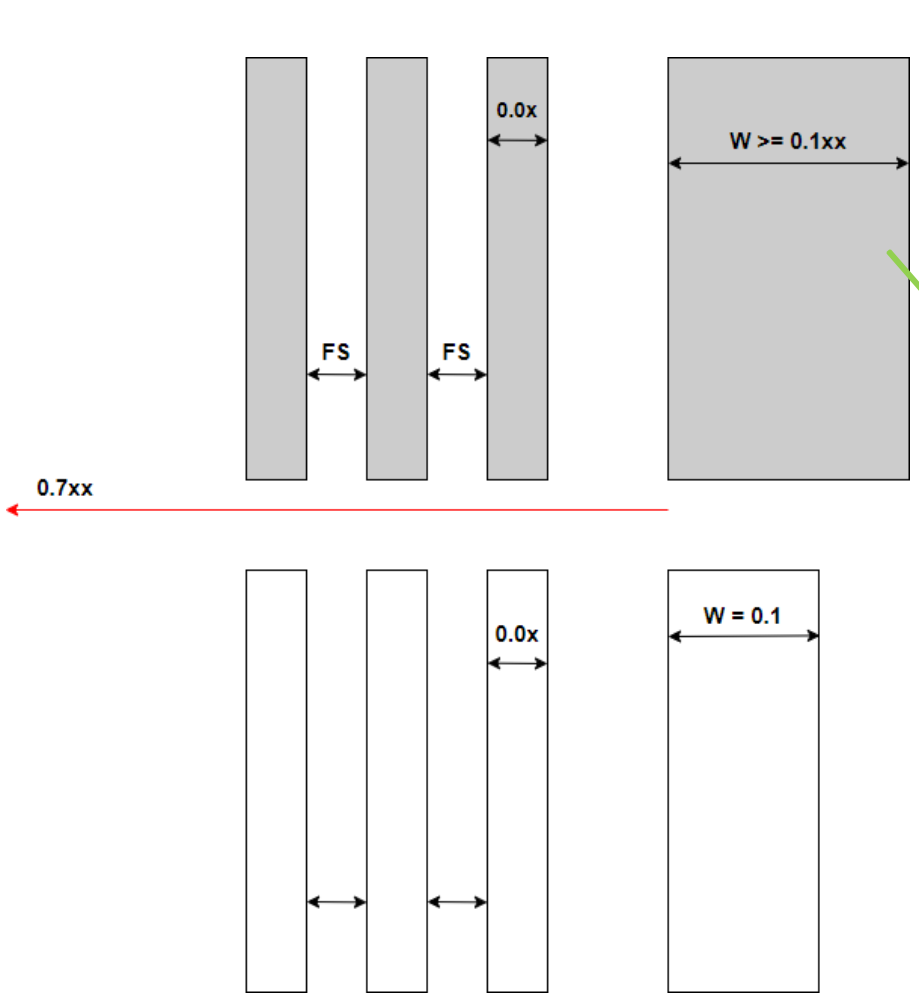
3. innovus > source lab_script/shrink_via.tcl
   補Follow pin連接到的via

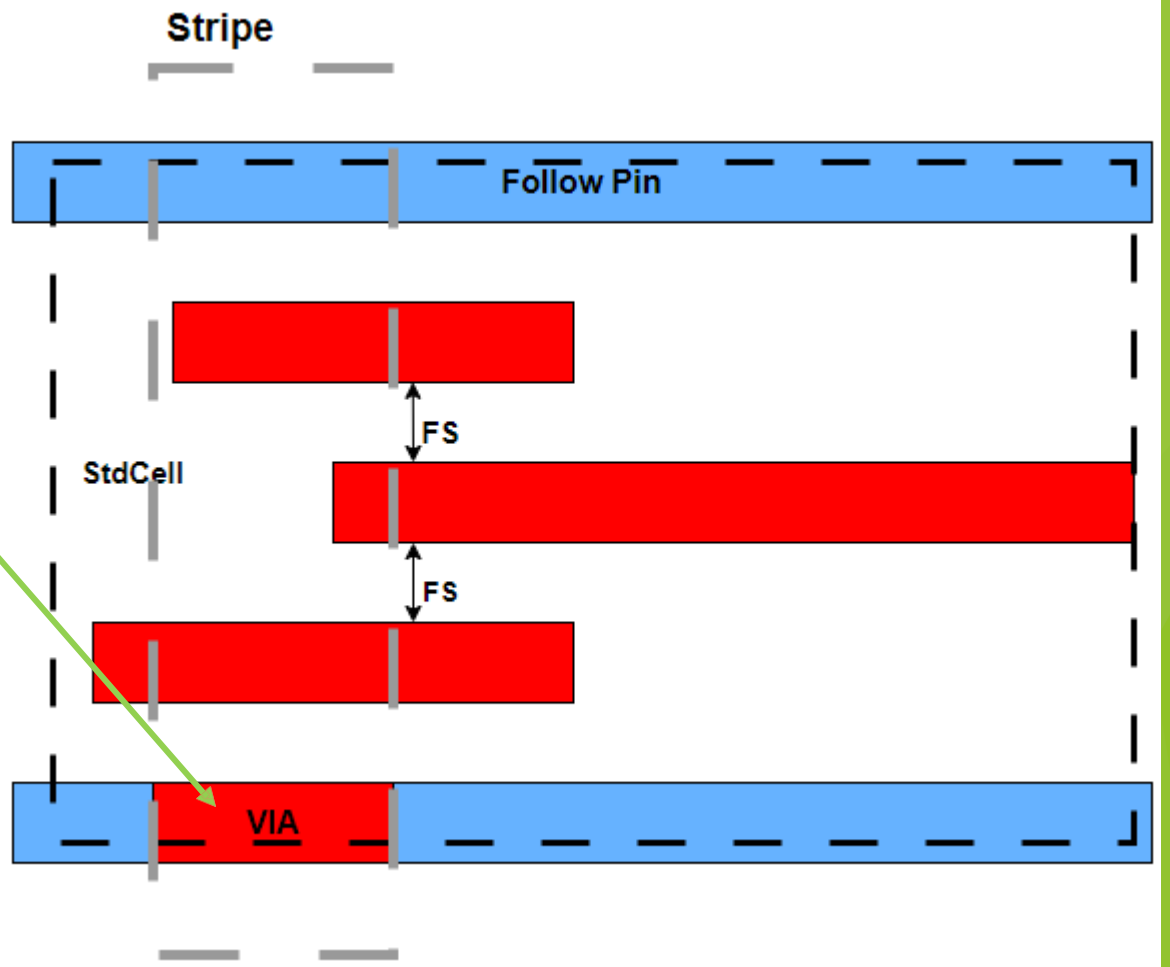




Power Route(4/8)

- Mxa Forbidden Spacing

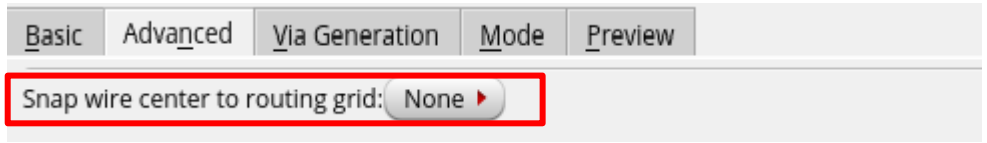


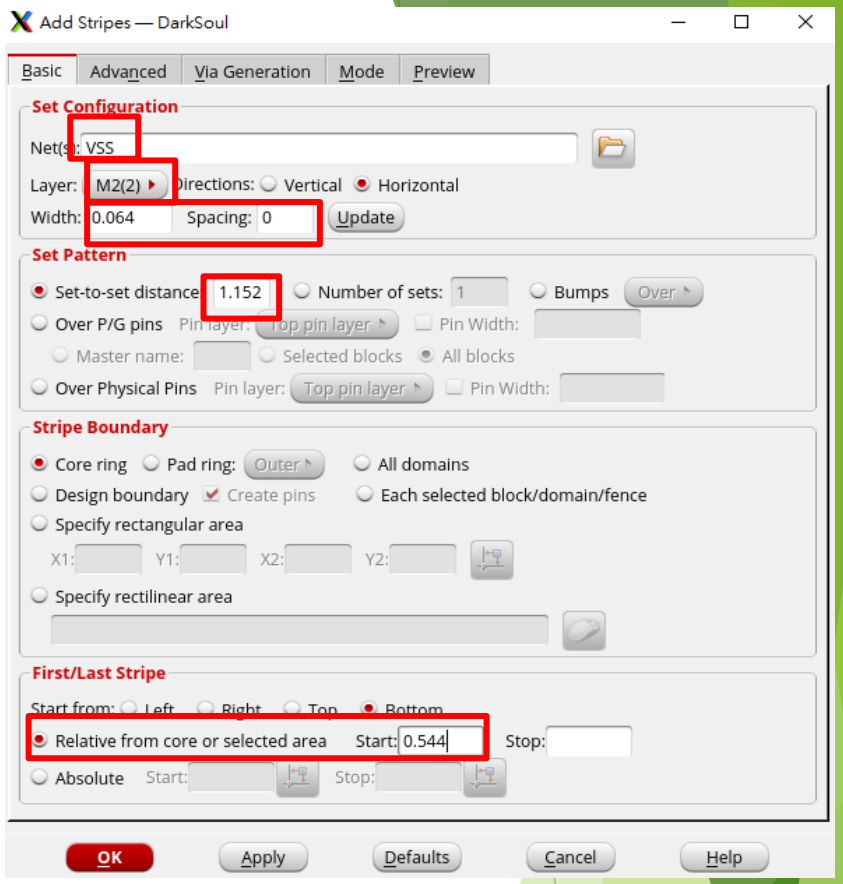


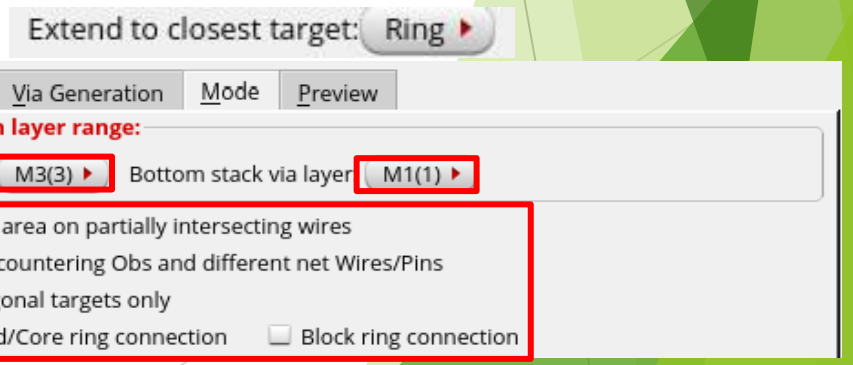
Power Route(5/8)

4. Power → Power Planning → Add Stripe → Basic
   - Nets: VSS
   - Layers: METAL2 and select Horizontal;
   - Set Width to 0.064 and Spacing to 0
   - In Set Pattern, select Set-to-set distance and input 1.152
   - Select Relative from core or selected area
   - Start: 0.544
5. Change to Advanced Page

   -Snap wire center to routing grid : select "None"
6. Change to Mode
   - Extend to closest target: Ring
   - Top Stack via layer: METAL3 Bottom Stack via layer: METAL1
   - un-choose Pad/Core ring connection & Block ring connection
7. Click Apply

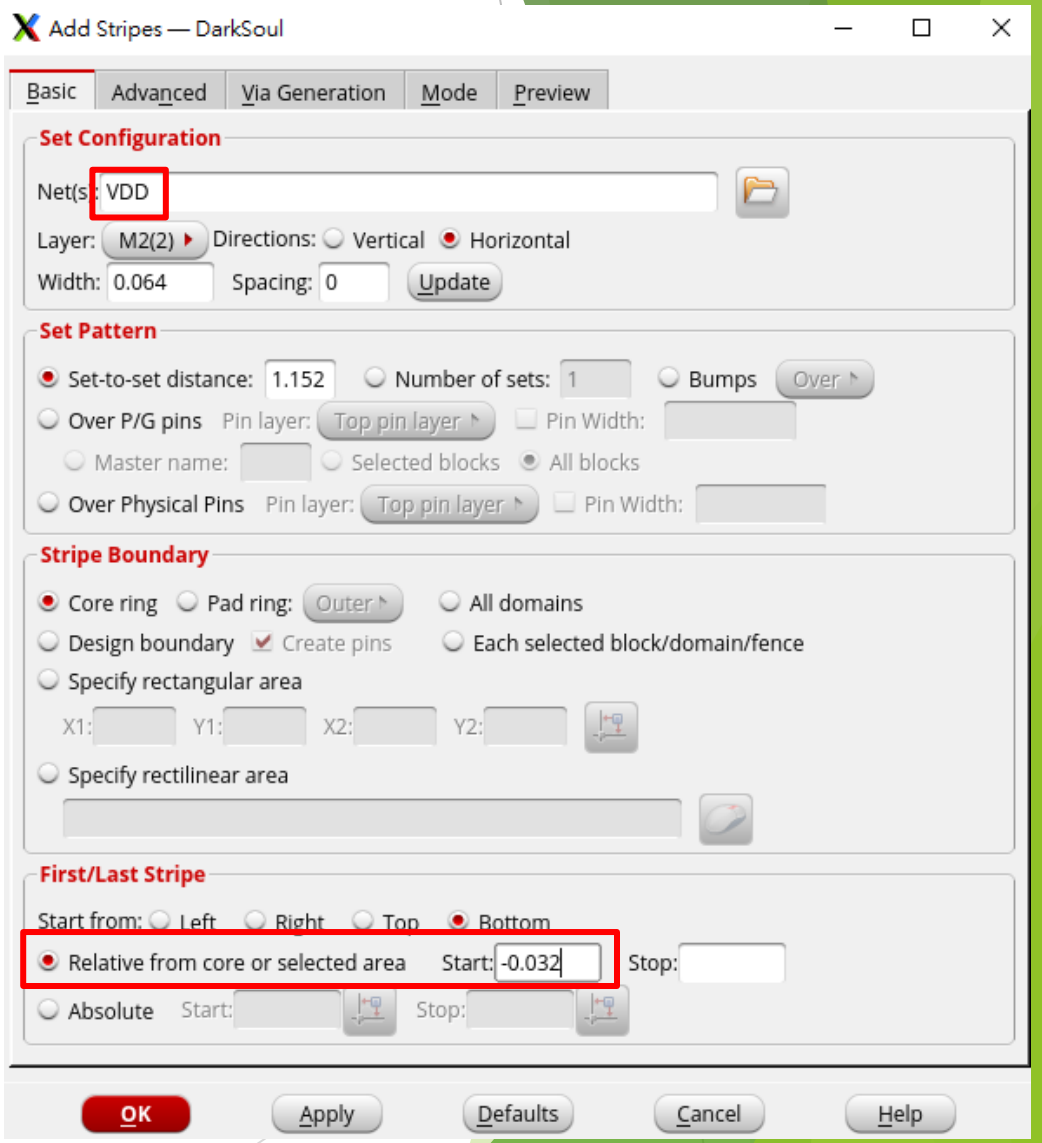






Power Route(6/8)

8. Change VSS to VDD
   - Nets: VDD
   - Start: -0.032
9. Click Apply

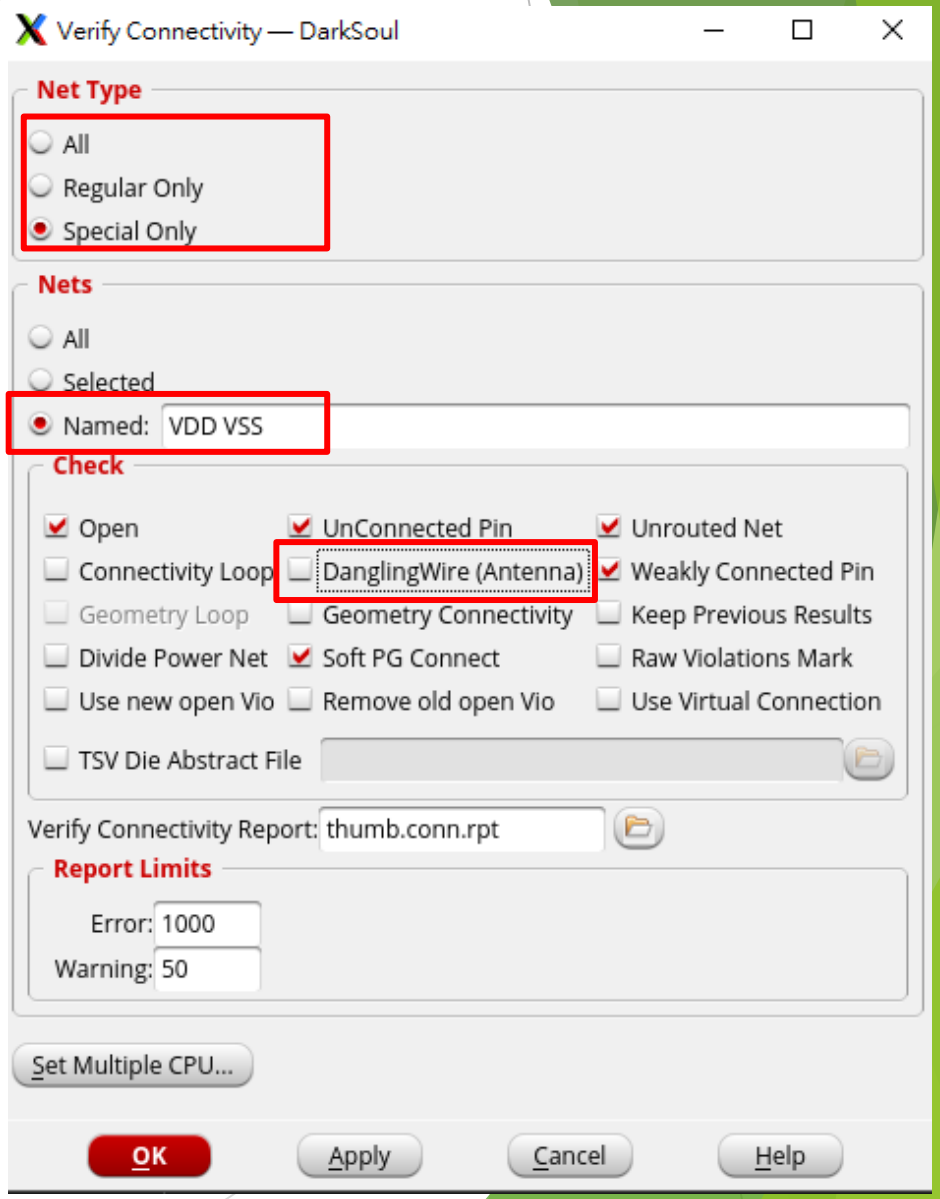


Power Route(7/8)

- **10. Check → Check Connectivity…**
  - **Net Type: Special Only**
  - **Nets: VDD VSS**
  - **In Check, un-choose DanglingWire (Antenna)**

There should not have any violation and X on the screen.

```
******** End: VERIFY CONNECTIVITY ********
  Verification Complete : 0 Viols.  0 Wrngs.
  (CPU Time: 0:00:00.1  MEM: 0.000M)
```

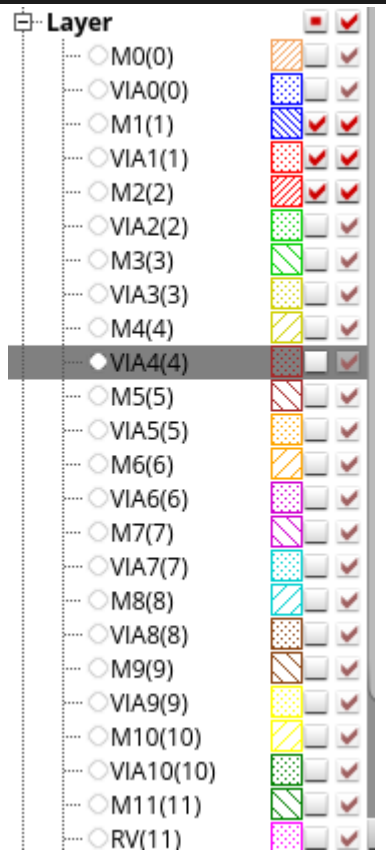


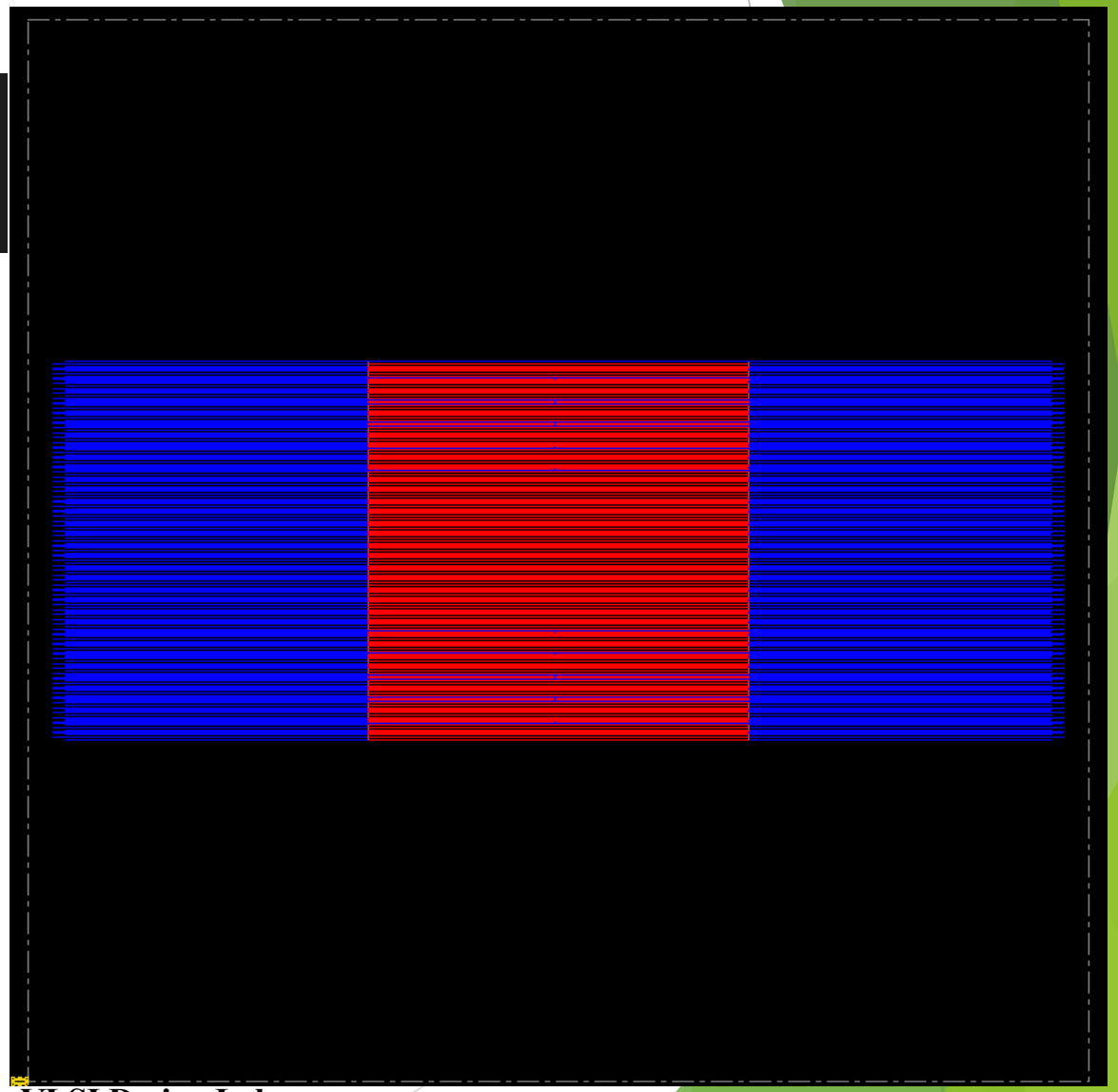
Power Route(8/8)

- 12. Check → CheckDRC… → OK

```
VERIFY DRC ..... Sub-Area : 80 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {222.208 222.208 248.940 248.736} 81 of 81
VERIFY DRC ..... Sub-Area : 81 complete 0 Viols.

Verification Complete : 0 Viols.
```



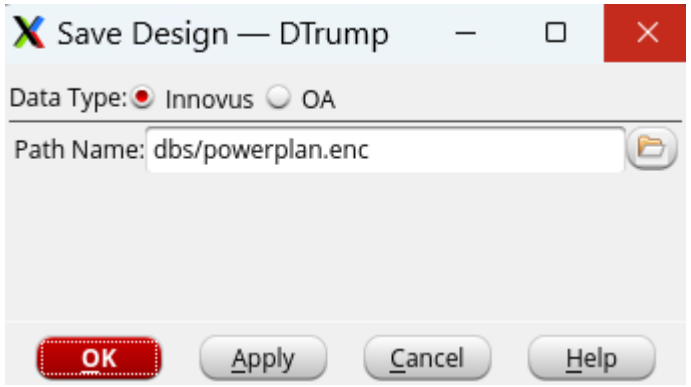


Power Route

- **Save Design**

  File →Save Design

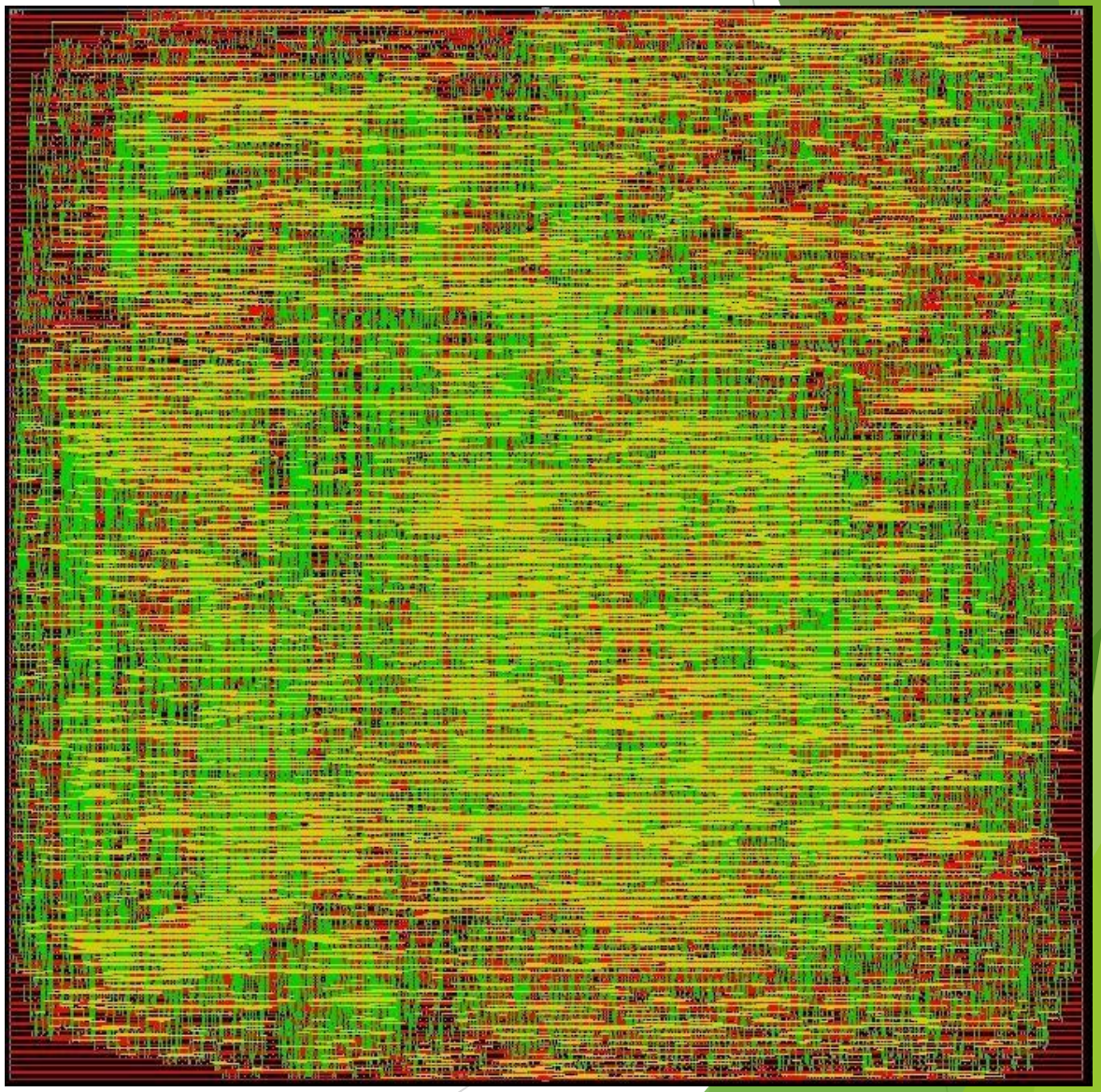
  dbs/powerplan.enc



Placement(1/3)

1. innovus > source lab_script/add_well_taps.tcl
2. Place → Place Standard Cell → Click OK
3. Place → Check Placement → Click Ok

```
Begin checking placement ... (start mem=4451.4M, init mem=4451.4M)
*info: Placed = 13587           (Fixed = 78)
*info: Unplaced = 0
```



Placement(2/3)

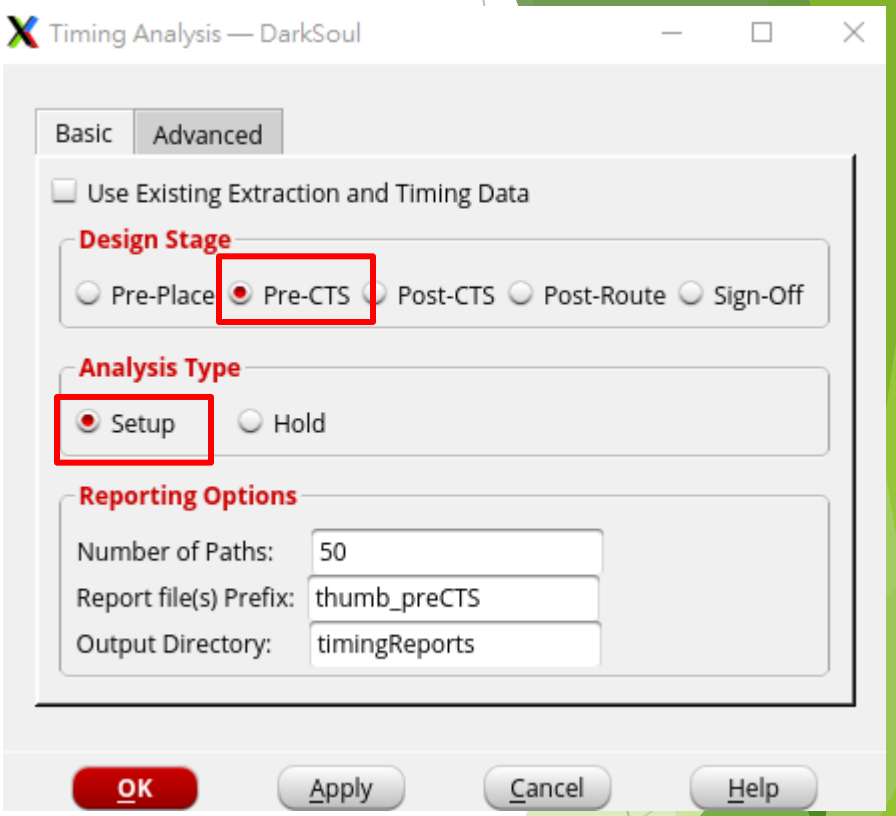
- **4. Timing → Report Timing**

  - Design Stage : Choose Pre-CTS

  - Analysis Type : Choose Setup

Setup mode	all	reg2reg	in2reg	reg2out	in2out	default
WNS (ns):	-1.009	-1.009	0.154	0.456	N/A	0.000
TNS (ns):	-164.026	-164.026	0.000	0.000	N/A	0.000
Violating Paths:	339	339	0	0	N/A	0
All Paths:	1351	735	1007	66	N/A	0

如果WNS有負的，照下一頁講義修



Placement(3/3)

5. ECO → Optimize Design

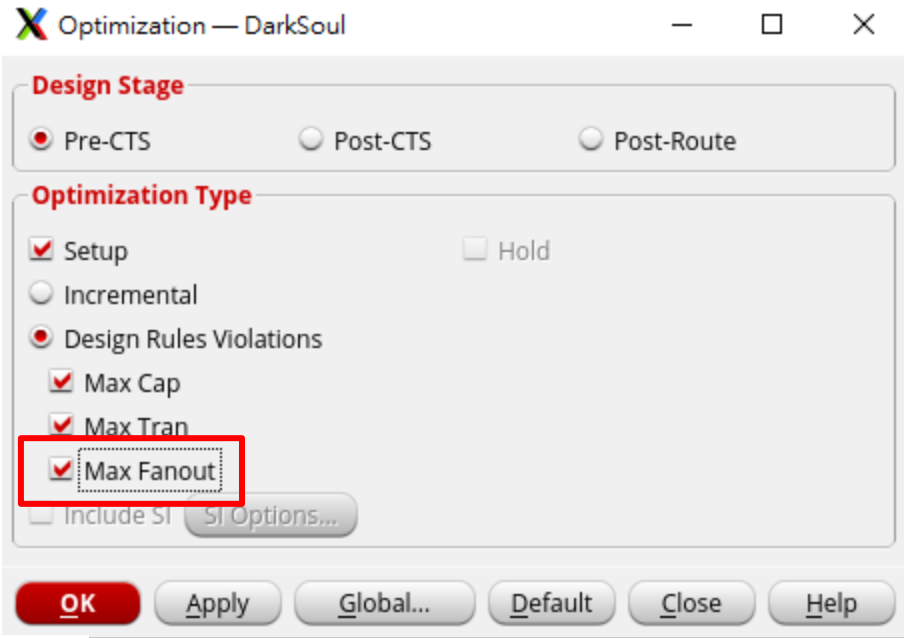
   - Design Stage : Choose Pre-CTS

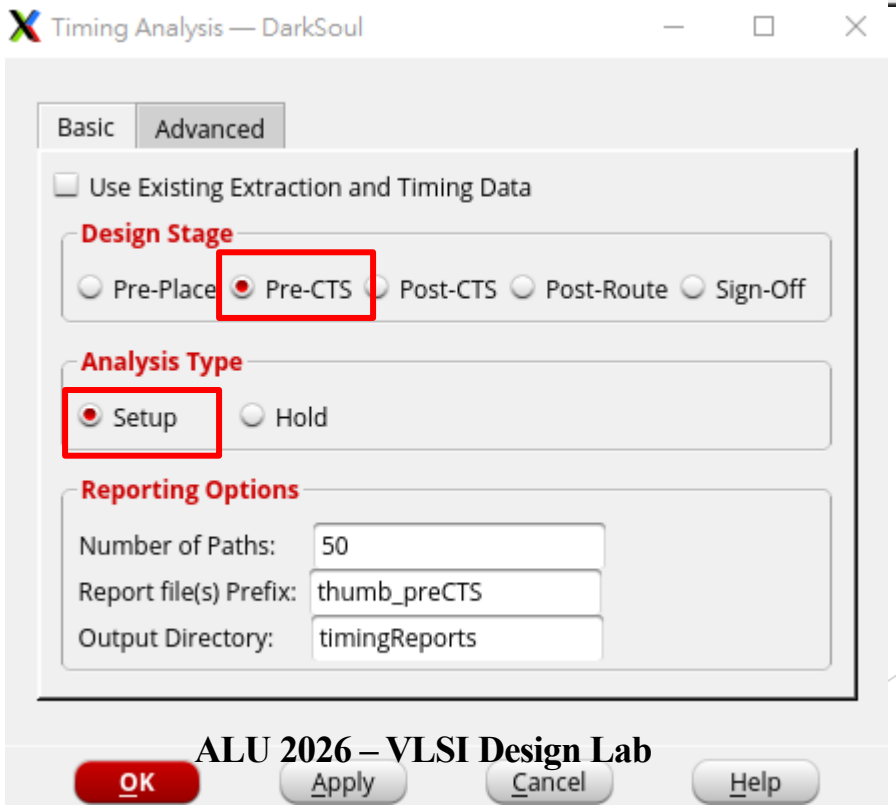
   - Optimization Type : Choose Max Fanout

6. Timing → Report Timing

   - Design Stage : Choose Pre-CTS

   - Analysis Type : Choose Setup



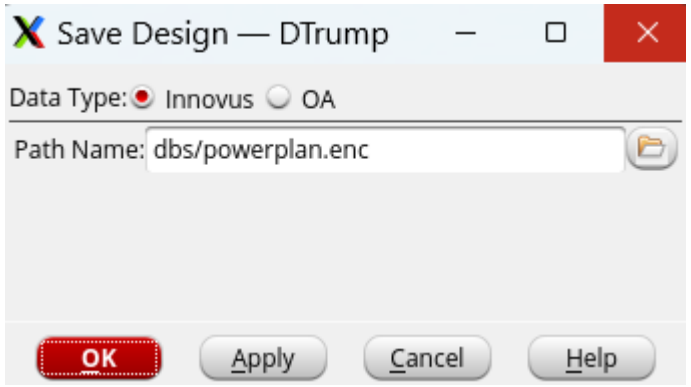


Placement

- **Save Design**

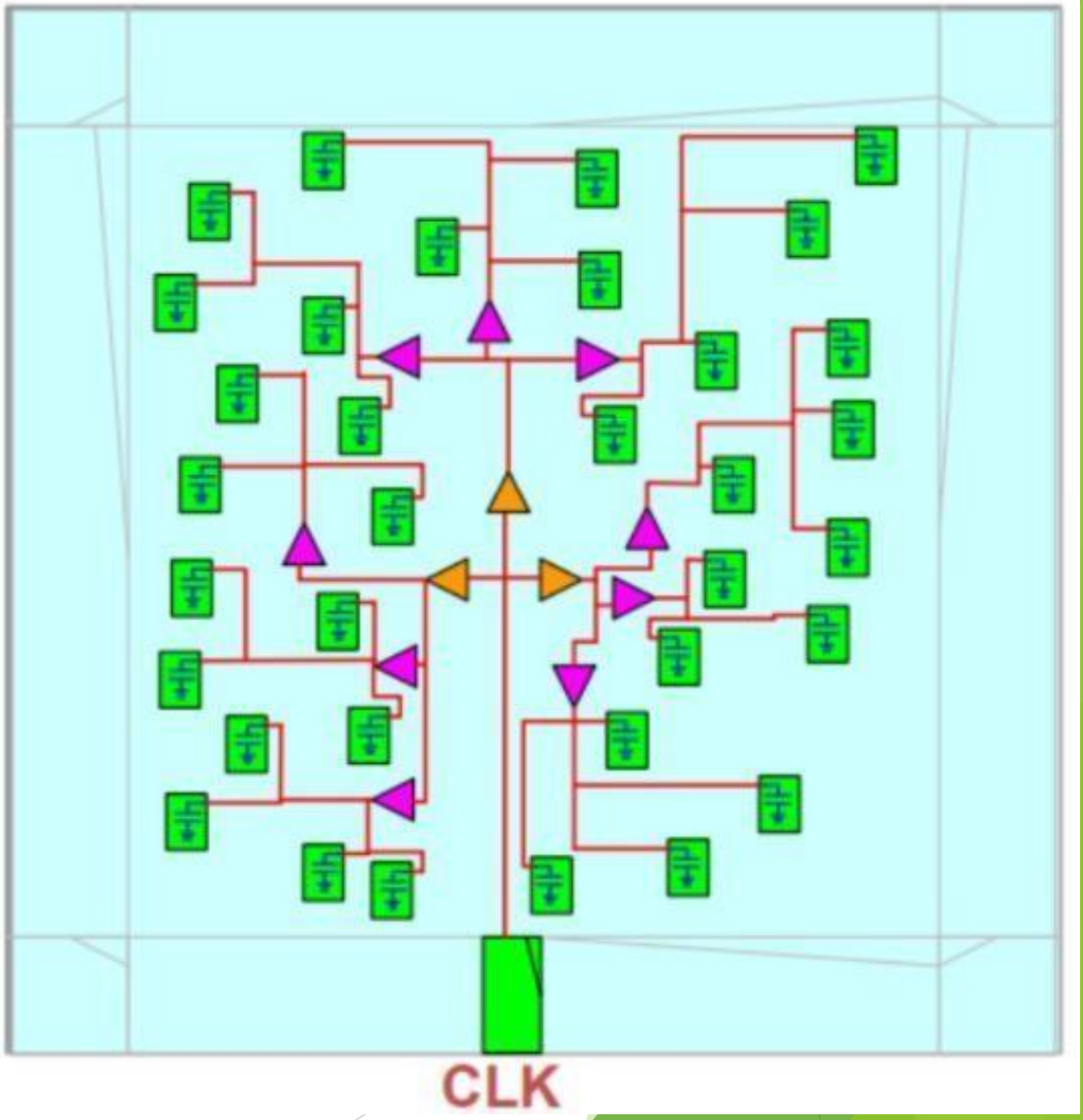
  File →Save Design

  dbs/placement.enc



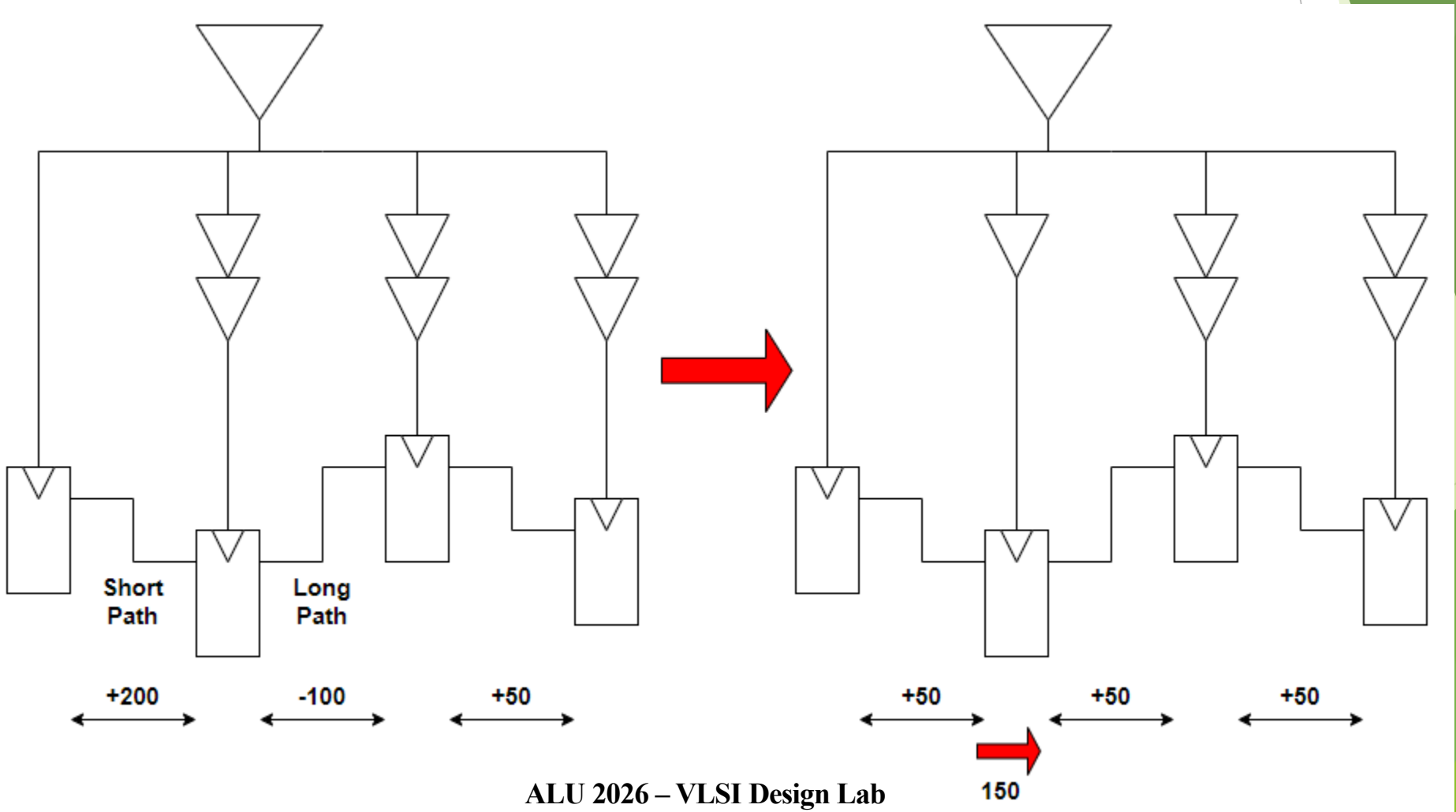
CTS(1/6)

- Clock Tree Synthesis(CTS), To solve Clock problem:
  - Heavy clock net loading
  - Long clock insertion delay
  - Clock skew
  - Skew across clocks
  - Clock to signal coupling effect
  - Clock is power hungry



CTS(2/6)

- Clock Concurrent Optimization



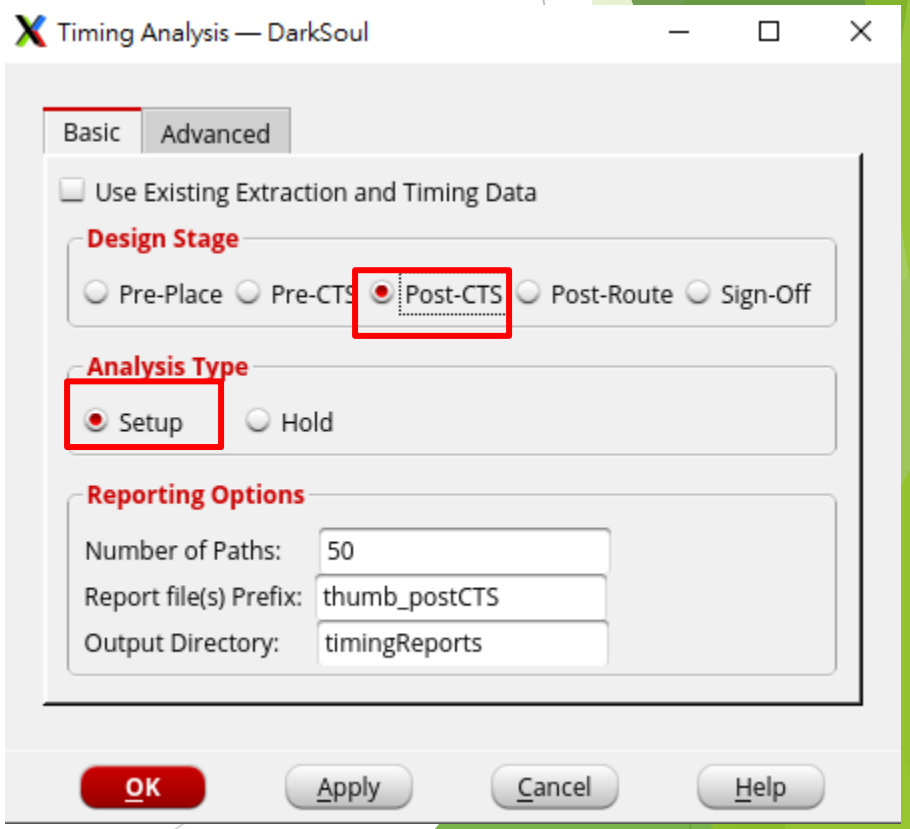
CTS(3/6)

1. innovus > set_interactive_constraint_modes [all_constraint_modes]
2. innovus > reset_clock_latency [all_clocks]
3. innovus > ccopt_design
4. Timing → Report Timing

   - Design Stage : Choose Post-CTS

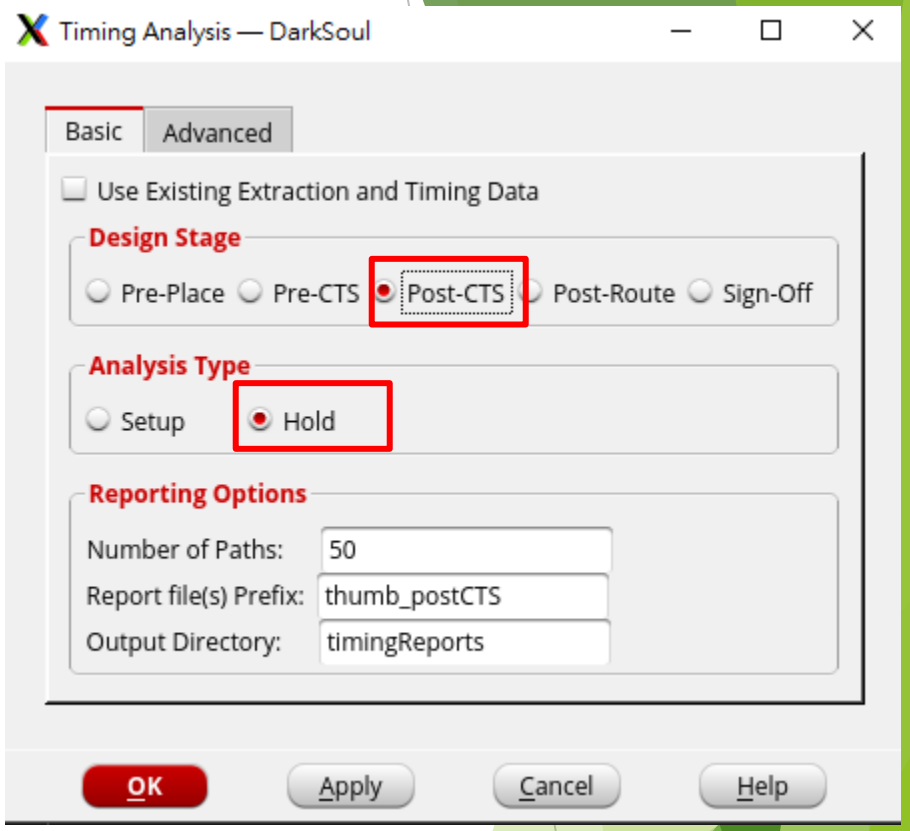
   - Analysis Type : Choose Setup

Setup mode	all	reg2reg	in2reg	reg2out	in2out	default
WNS (ns):	0.001	0.001	0.297	0.413	N/A	0.000
TNS (ns):	0.000	0.000	0.000	0.000	N/A	0.000
Violating Paths:	0	0	0	0	N/A	0
All Paths:	1349	734	1006	66	N/A	0



CTS(4/6)

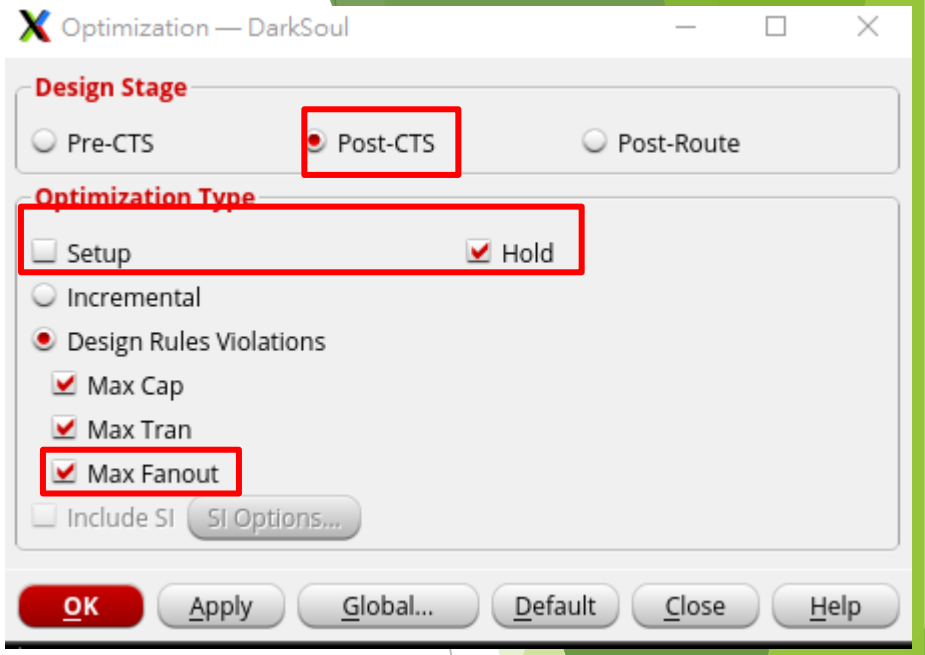
- **5. Timing → Report Timing**
  - **- Design Stage : Choose Post-CTS**
  - **- Analysis Type : Choose hold**

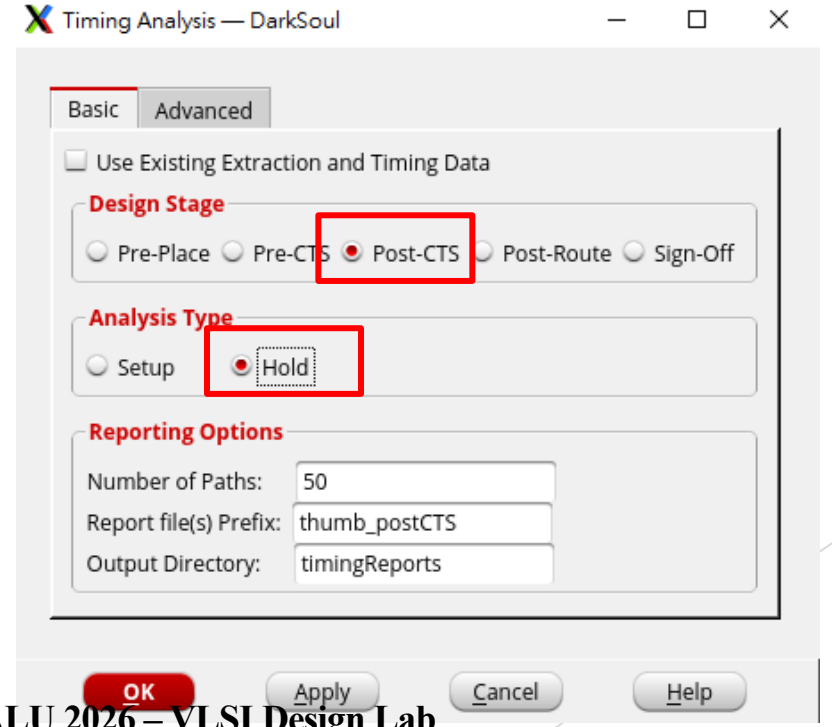


Hold mode	all	reg2reg	in2reg	reg2out	in2out	default
WNS (ns):	-0.061	0.027	-0.061	0.041	N/A	0.000
TNS (ns):	-28.644	0.000	-28.644	0.000	N/A	0.000
Violating Paths:	724	0	724	0	N/A	0
All Paths:	1349	734	1006	66	N/A	0

CTS(5/6)

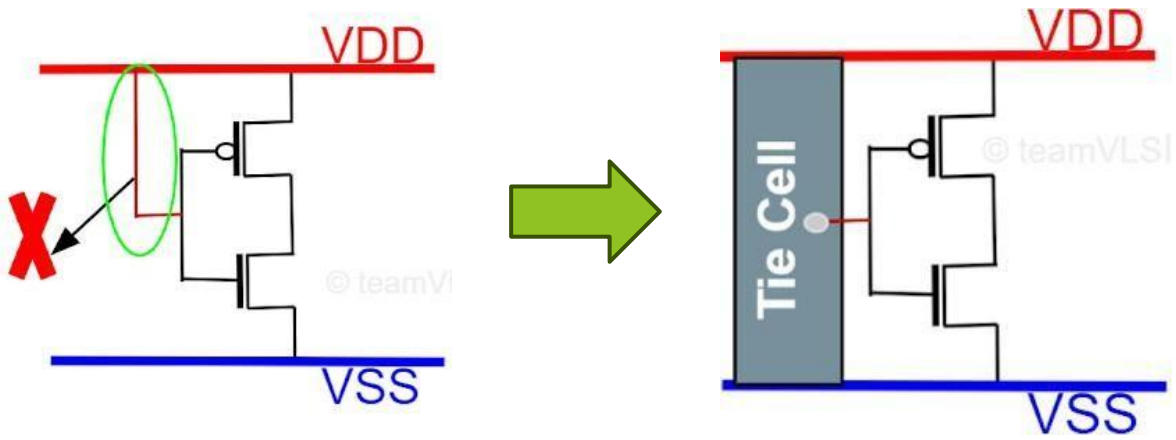
6. ECO → Optimize Design
   - Design Stage : Choose Post-CTS
   - Optimization Type : Choose Hold
   - Optimization Type : Choose Max Fanout
7. Timing → Report Timing
   - Design Stage : Choose Post-CTS
   - Analysis Type : Choose Hold

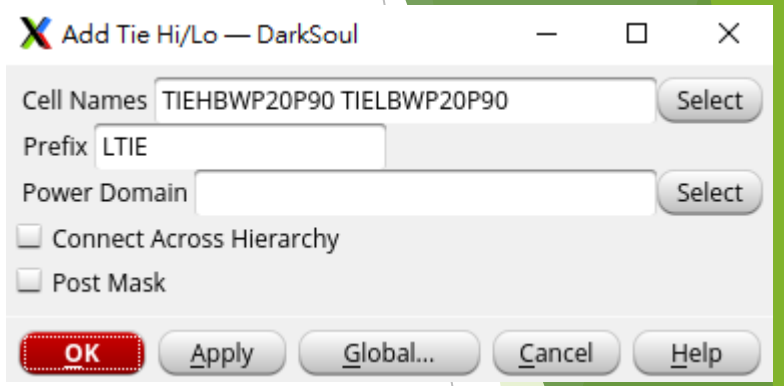


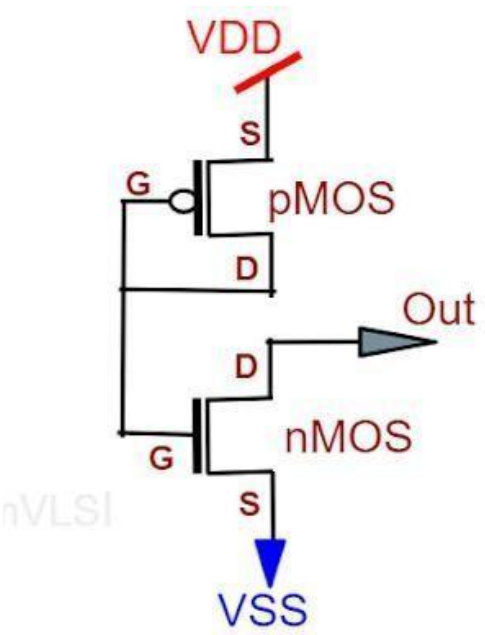


CTS(6/6)

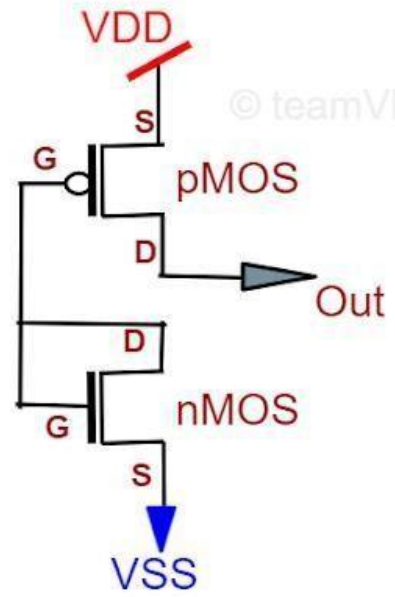
- **8. Add Tie HI / LO cell**
  - **- Place → Tie HI / LO → Add**
  - **- Clcik Ok**







Tie-low Cell



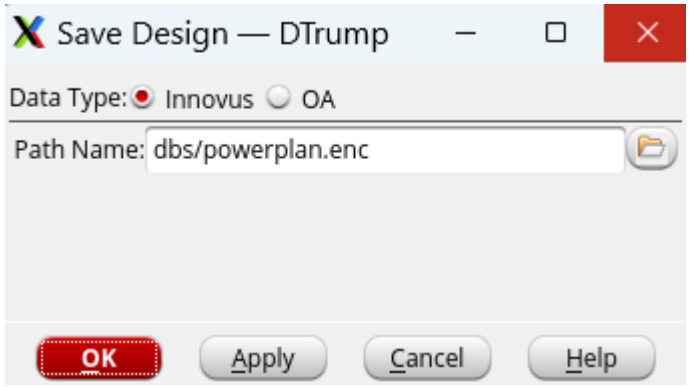
Tie-high Cell

CTS

- **Save Design**

  File →Save Design

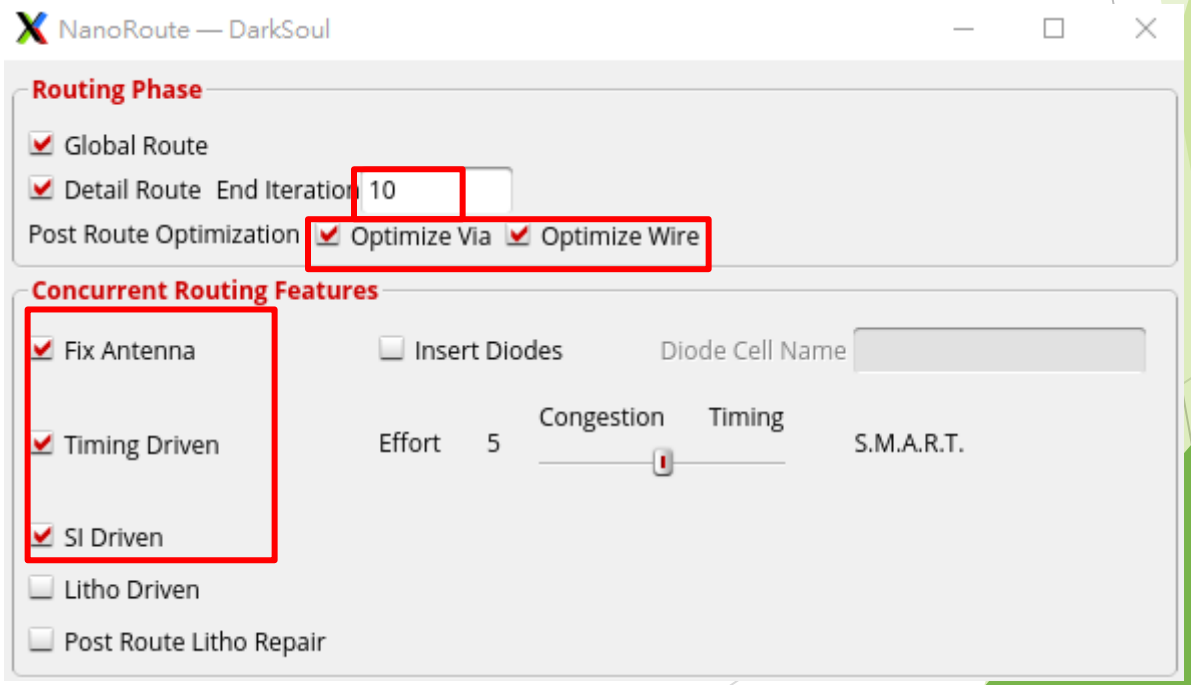
  dbs/cts.enc



Route(1/5)

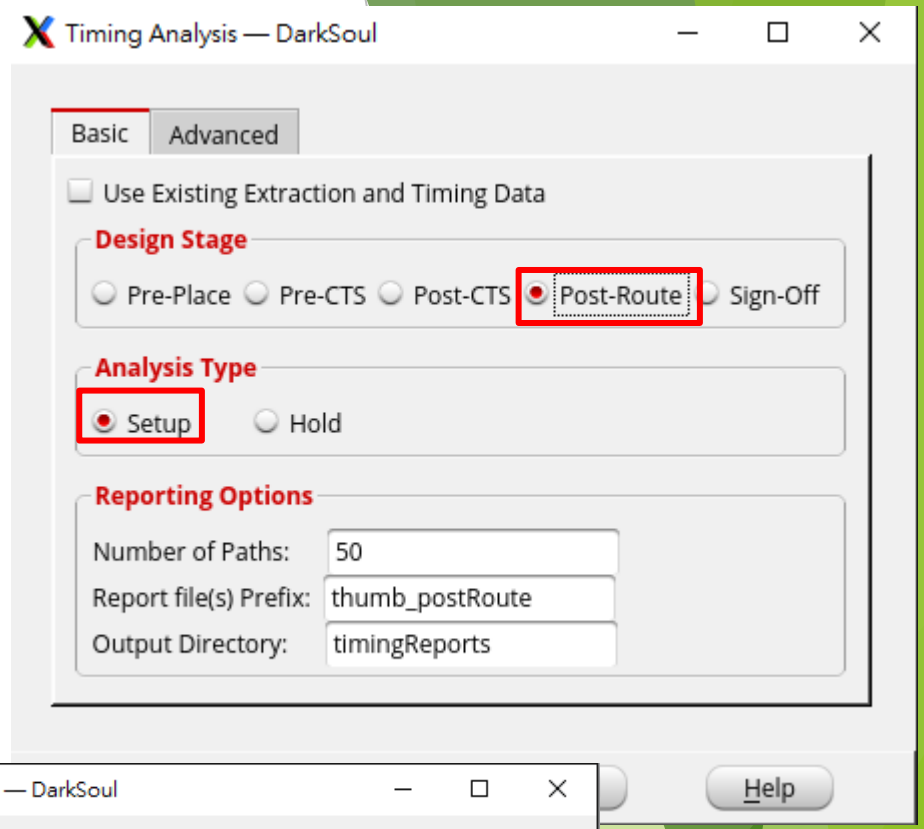
1. **Start routing:**

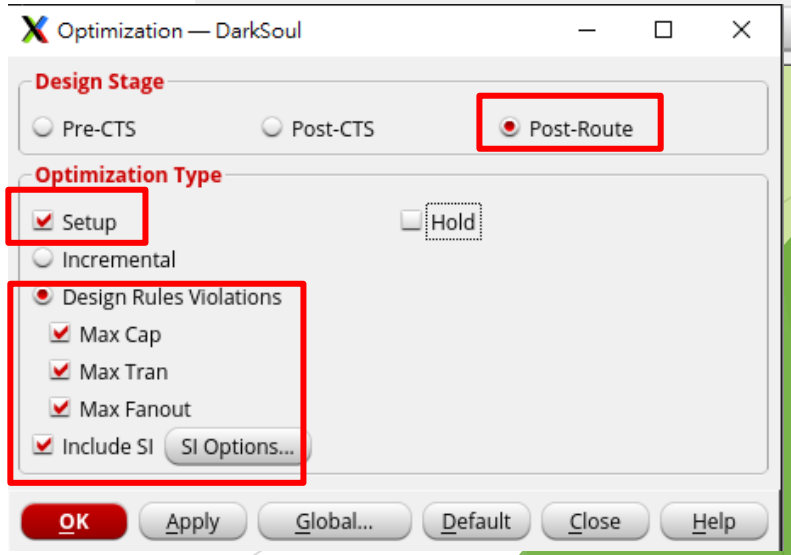
   - **Route → NanoRoute → Route**
   - **Detail Route End Iteration :10**
   - **In Routing Phase, select Optimize Via & Optimize Wie**
   - **In Concurrent Routing Features, select Fix Antenna, Timing Driven, SI Driven**
   - **Click OK**



Route(2/5)

- 2. innovus > set_db delaycal_enable_si true
- 3. Timing → Report Timing
  - Design Stage : Choose Post-Route
  - Analysis Type : Choose Setup
- 4. ECO → Optimize Design
  - Design Stage : Choose Post-Route
  - Optimization Type : Choose Setup
  - Click OK

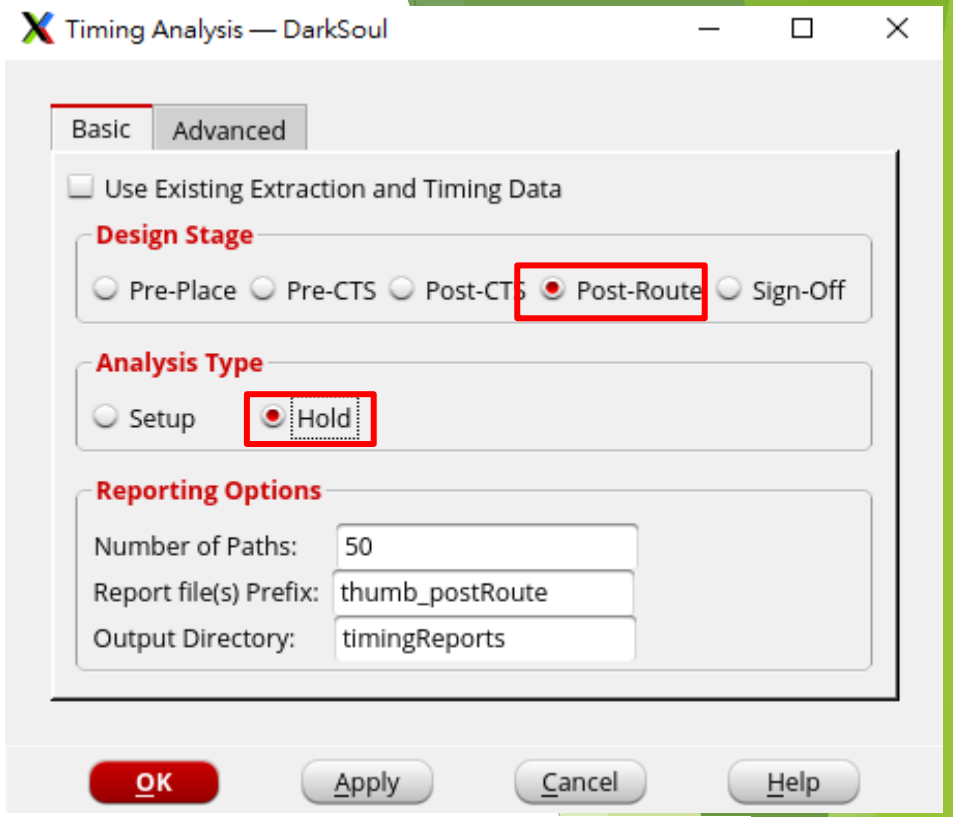




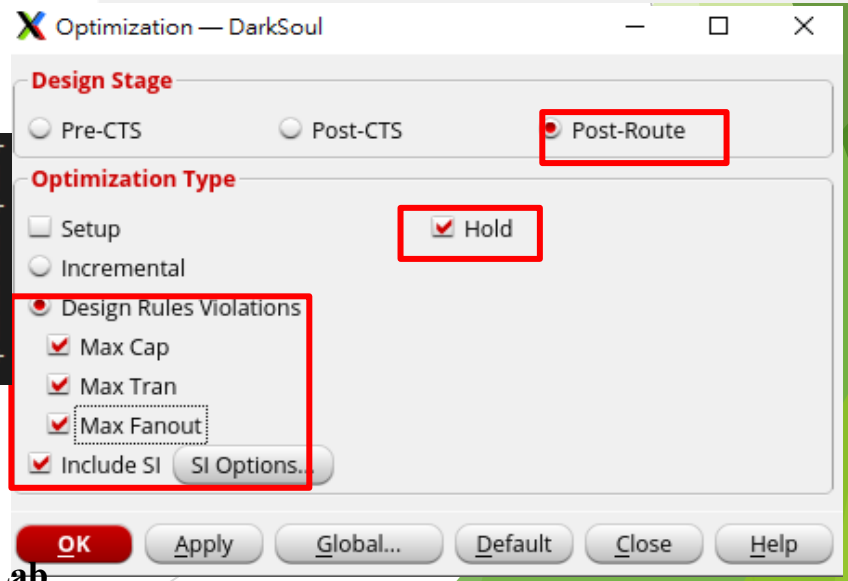
Setup mode	all	reg2reg	in2reg	reg2out	in2out	default
WNS (ns):	0.398	0.746	0.398	2.303	N/A	0.000
TNS (ns):	0.000	0.000	0.000	0.000	N/A	0.000
Violating Paths:	0	0	0	0	N/A	0
All Paths:	1351	735	1007	66	N/A	0

Route(3/5)

- 5. Timing → Report Timing
  - Design Stage : Choose Post-Route
  - Analysis Type : Choose Hold
- 6. ECO → Optimize Design
  - Design Stage : Choose Post-Route
  - Optimization Type : Choose Hold
  - Click OK



Hold mode	all	reg2reg	in2reg	reg2out	in2out	default
WNS (ns):	0.004	0.004	0.034	0.086	N/A	0.000
TNS (ns):	0.000	0.000	0.000	0.000	N/A	0.000
Violating Paths:	0	0	0	0	N/A	0
All Paths:	1351	735	1007	66	N/A	0



Route(4/5)

7. Timing → Report Timing

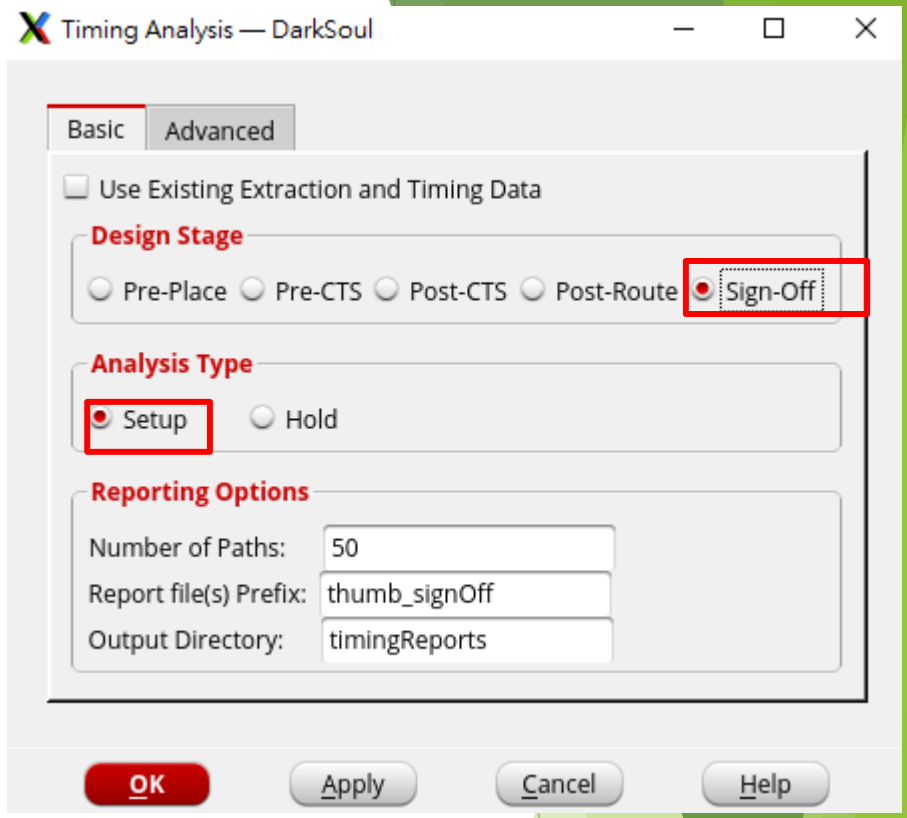
   - Design Stage : Choose Sign-Off

   - Analysis Type : Choose Setup

8. innovus > set_multi_cpu_usage -remote_host 8

   innovus > opt_signoff –setup

   (如果Timing沒過再修)



Setup mode	all	reg2reg	in2reg	reg2out	in2out	default
WNS (ns):	0.730	0.730	1.319	2.302	N/A	0.000
TNS (ns):	0.000	0.000	0.000	0.000	N/A	0.000
Violating Paths:	0	0	0	0	N/A	0
All Paths:	1351	735	1007	66	N/A	0

Route(5/5)

- 9. Timing → Report Timing

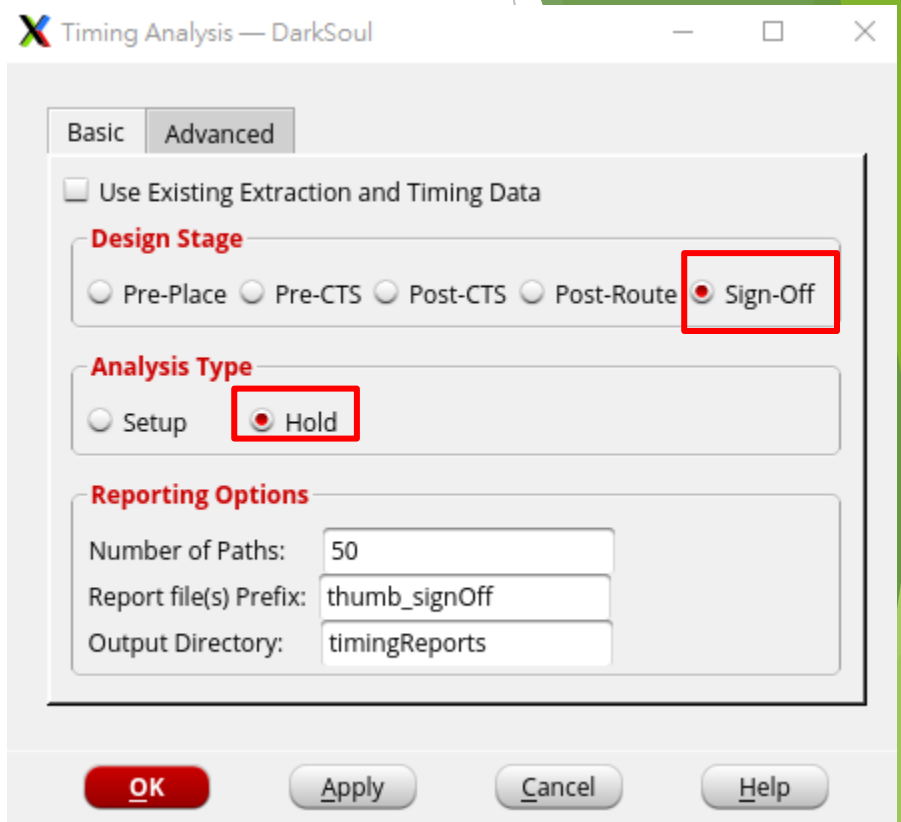
  - Design Stage : Choose Sign-Off

  - Analysis Type : Choose Hold

- 10. innovus > set_multi_cpu_usage –remote_host 8

  innovus > opt_signoff –hold

  (如果Timing沒過再修)



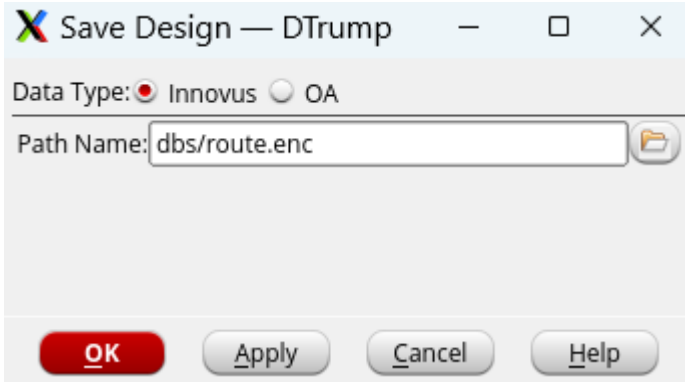
Hold mode	all	reg2reg	in2reg	reg2out	in2out	default
WNS (ns):	0.008	0.008	0.018	0.090	N/A	0.000
TNS (ns):	0.000	0.000	0.000	0.000	N/A	0.000
Violating Paths:	0	0	0	0	N/A	0
All Paths:	1351	735	1007	66	N/A	0

Route

- **Save Design**

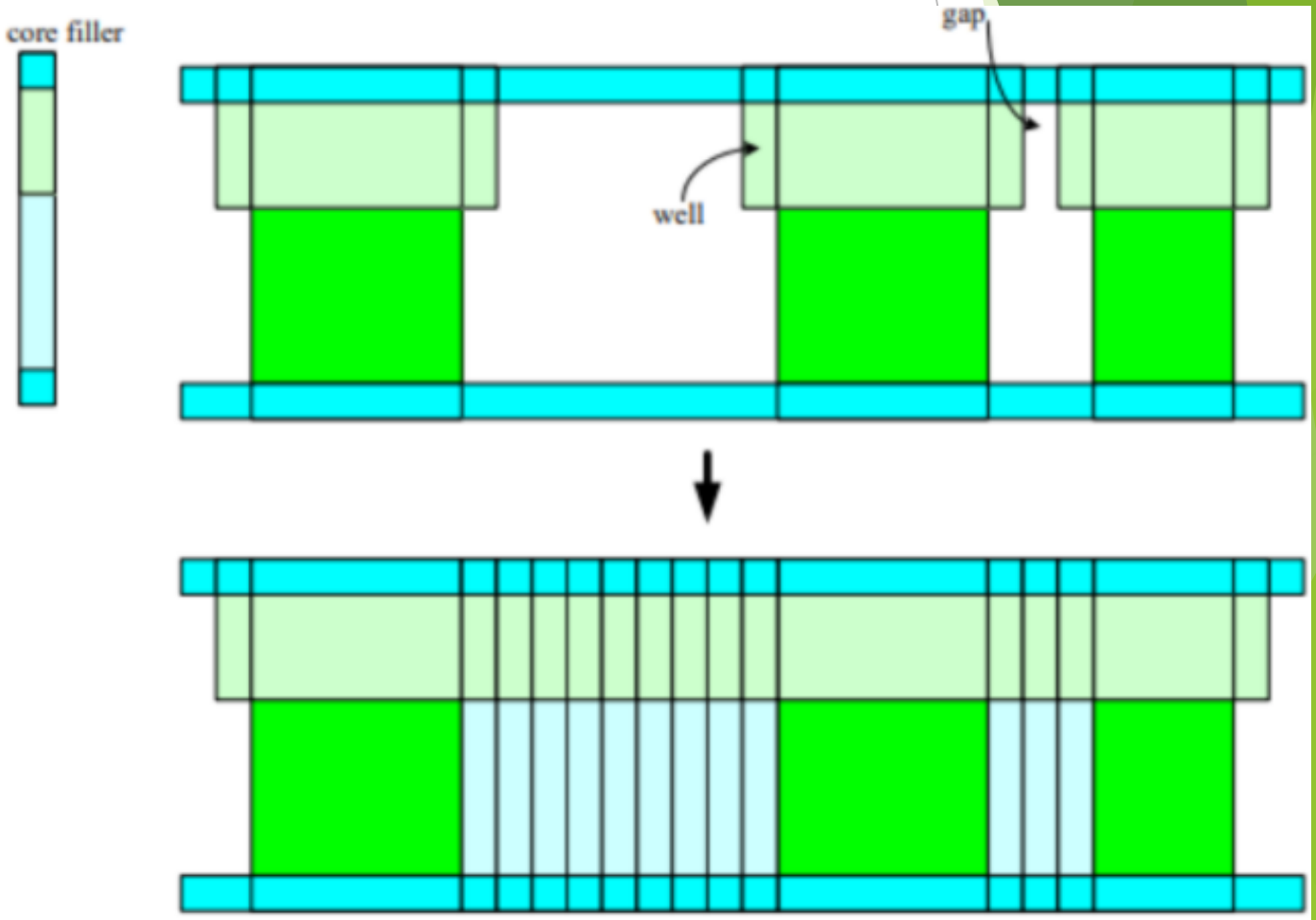
  File →Save Design

  dbs/route.enc



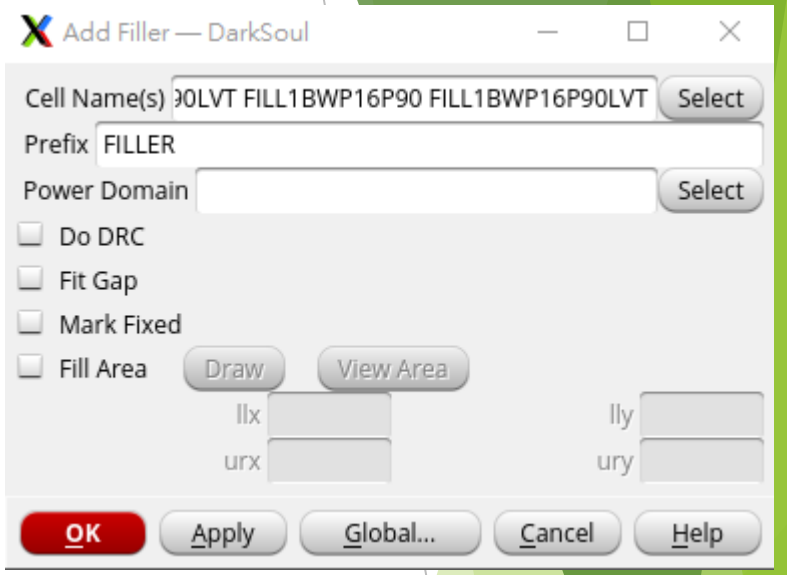
Post processing(1/5)

- 1. Add Core Filler
  - To fill the gap between cells.
  - Start from wider filler to narrow filler



Post processing(2/5)

- 2. Add Core Filler:
  - innovus > get_db add_fillers_cells
  - Place →Physical Cells → Add Filler → Click OK
- 3. Add VIA1 between M1 and M2
  - innovus > source lab_script/post_via_drop.tcl
- 4. Check DRC
  - innovus > set_db check_drc_limit 10000
  - innovus > check_drc
  - innovus > source lab_script/fix_manual_via12.tcl

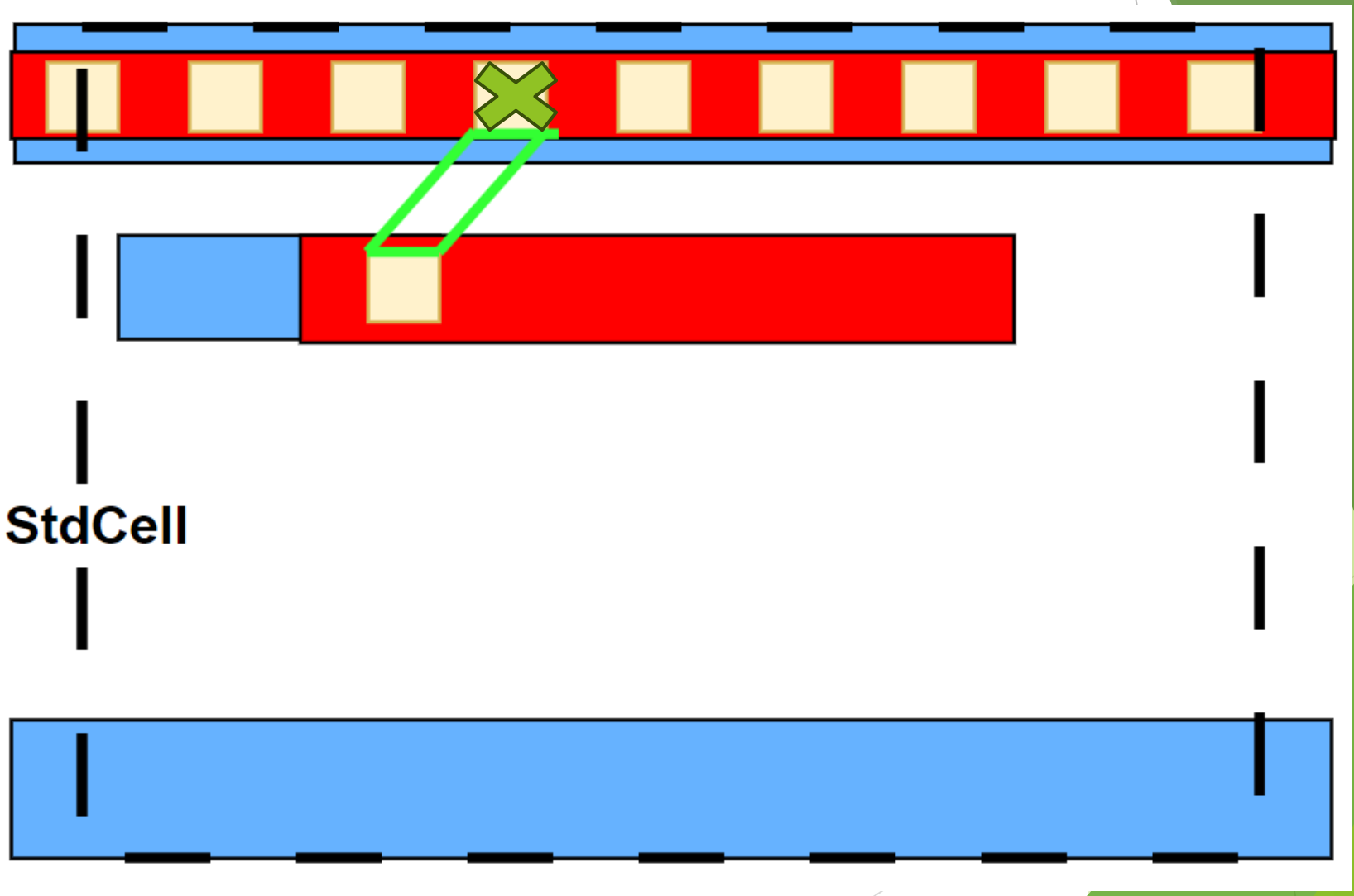



```
VERIFY DRC ..... Sub-Area: {0.000 187.488 27.280 214.272} 64 of 81  Thread : 6
VERIFY DRC ..... Thread : 6 finished.
VERIFY DRC ..... Sub-Area: {54.560 187.488 81.840 214.272} 66 of 81  Thread : 7
VERIFY DRC ..... Thread : 7 finished.
VERIFY DRC ..... Sub-Area: {218.240 187.488 241.380 214.272} 72 of 81  Thread : 0
VERIFY DRC ..... Thread : 0 finished.
VERIFY DRC ..... Sub-Area: {163.680 187.488 190.960 214.272} 70 of 81  Thread : 4
VERIFY DRC ..... Thread : 4 finished.

 Verification Complete : 3 Viols
```

Post processing(3/5)

fix_manual_via12.tcl



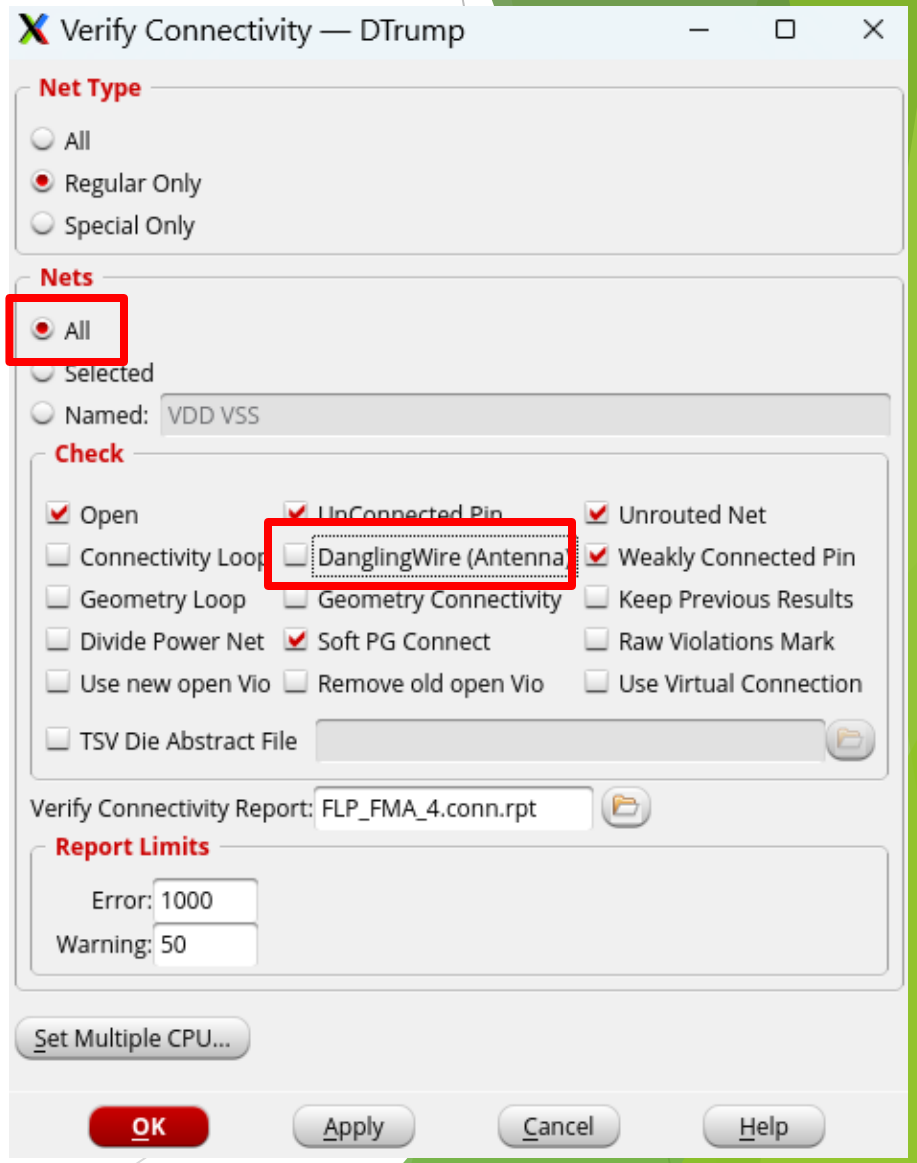
Post processing(4/5)

- 5. Check → CheckDRC… → OK
- 6. Check →Check Connectivity
  - Net Type → Choose Regular Only
  - Net → All
  - Check → un-choose DanglingWire(Antenna)
- 7. Finish Design
  - innovus > source lab_script/write_netlist.tcl
  - innovus > source lab_APR_script/write_sdf.tcl
  - innovus > source lab_script/write_stream.tcl

```
VERIFY DRC ..... Thread : 4 finished.
VERIFY DRC ..... Thread : 6 finished.
VERIFY DRC ..... Sub-Area: {190.960 187.488 218.240 214.272} 71 of 81  Thread : 3
VERIFY DRC ..... Thread : 3 finished.

 Verification Complete : 0 Viols.

*** End Verify DRC (CPU: 0:01:18  ELAPSED TIME: 14.00  MEM: 8.0M) ***
```



Post processing(5/5)

write_netlist.tcl

```
delete_empty_hinsts

write_netlist outputs/FMA_pipe_pr.v
set TAP_CELL_LIST [get_db [get_db base_cells TAP_*] .name]
set BOUNDARY_CELL_LIST [get_db [get_db base_cells BOUNDARY_*] .name]
set DECAP_CELL_LIST [get_db [get_db base_cells DCAP_*] .name]
set PVDD_CELL_LIST [get_db [get_db base_cells PVDD*] .name]
set FILLER_CELL_LIST [get_db [get_db base_cells FILL*] .name]
set PFILLER_CELL_LIST [get_db [get_db base_cells PFILL*] .name]
set PCORNER_CELL_LIST [get_db [get_db base_cells PCORNER*] .name]
write_netlist -include_pg_ports  -include_phys_cells "$TAP_CELL_LIST $BOUNDARY_CELL_LIST $DECAP_CELL_LIST $PVDD_CELL_LIST $FILLER_CELL_LIST $PFILLER_CELL_LIST $PCORNER_CELL_LIST" outputs/FMA_pipe_pg.v
```

write_sdf.tcl

```
write_sdf outputs/FMA_pipe_ff0p88v125c.sdf -max_view AV_func_ff0p88v125c
write_sdf outputs/FMA_pipe_ss0p72vm40c.sdf -max_view AV_func_ss0p72vm40c
```

write_stream.tcl

```
write_stream outputs/FMA_pipe.gds -map_file stream_out_map -lib_name DesignLib \
      -merge { \
            /home/cad/CBDK/ADFP/Executable_Package/Collaterals/IP/stdcell/N16ADFP_StdCell/GDS/N16ADFP_StdCell.gds      \
      } \
      -uniquify_cell_names -unit 1000 -mode all -report_file write_stream.log

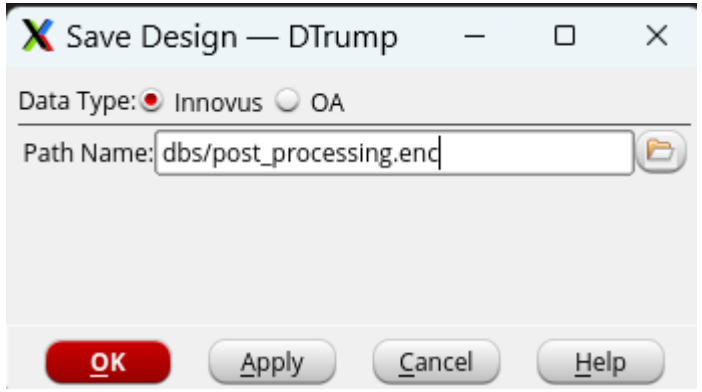
#create_pin_text -cells thumb label_loc.txt
```

Post processing

- **Save Design**

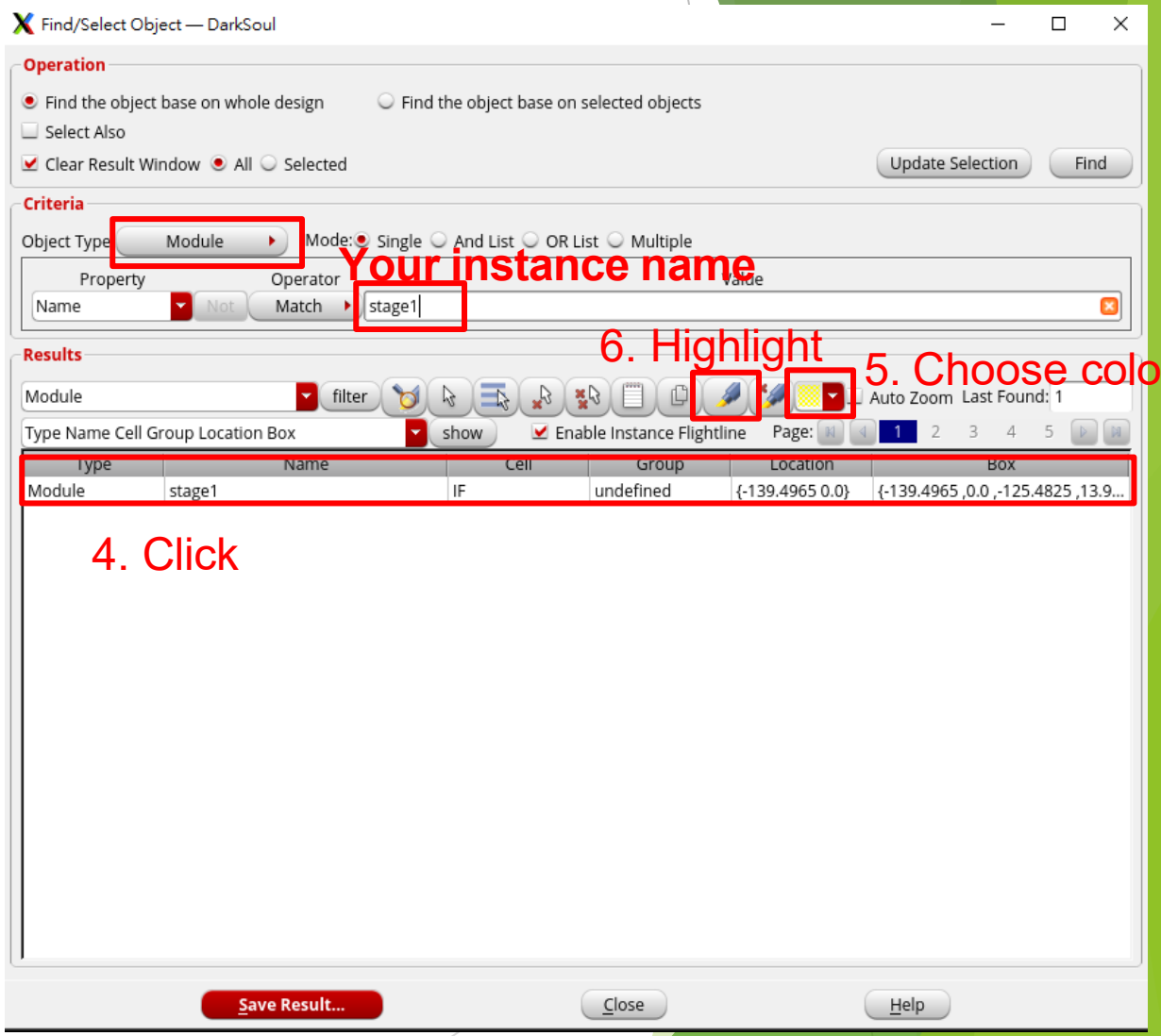
  File →Save Design

  dbs/post_processing.enc

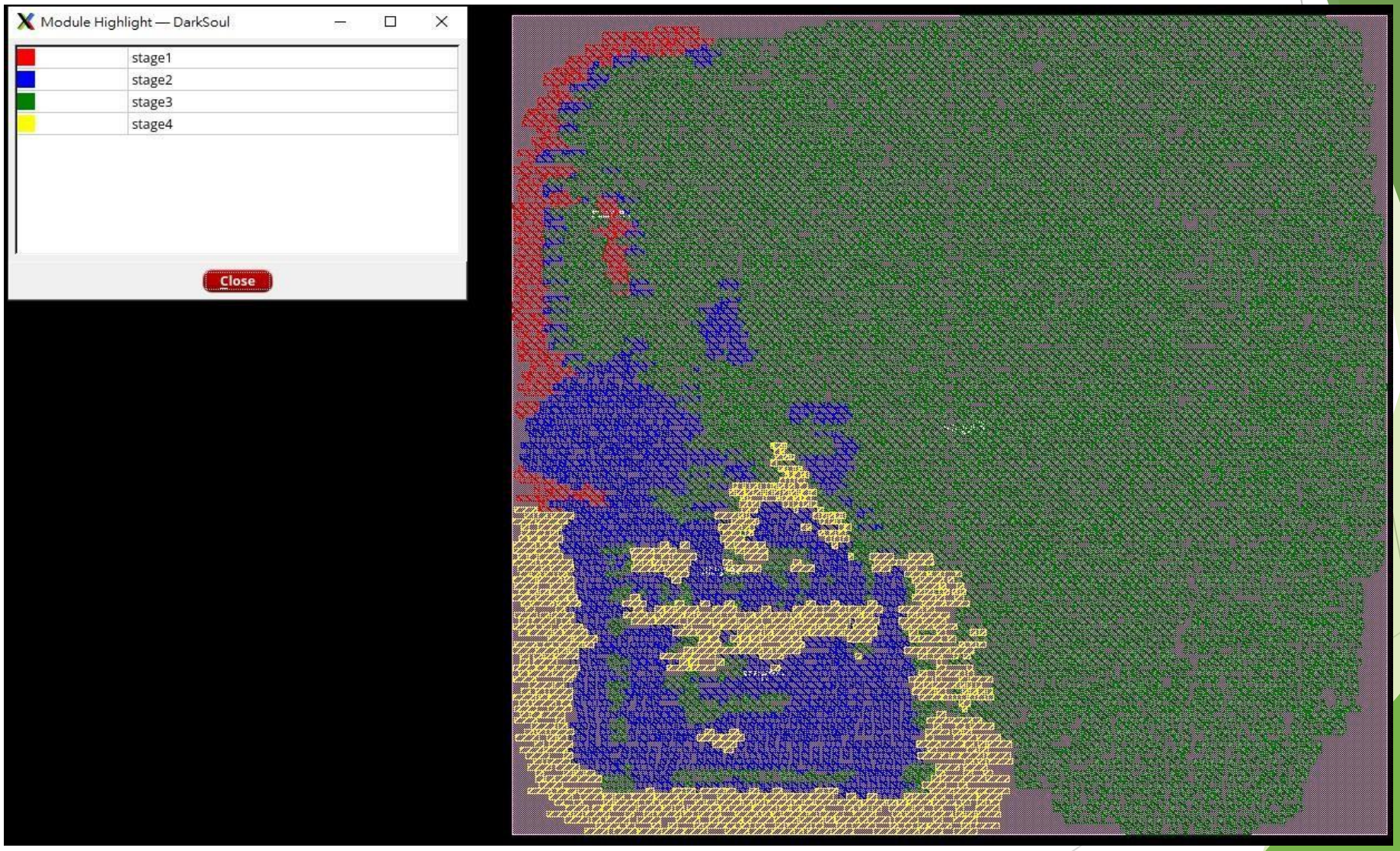


Highlight module(1/3)

1. ctrl +f
2. Select Module
3. Type stage1 → Enter



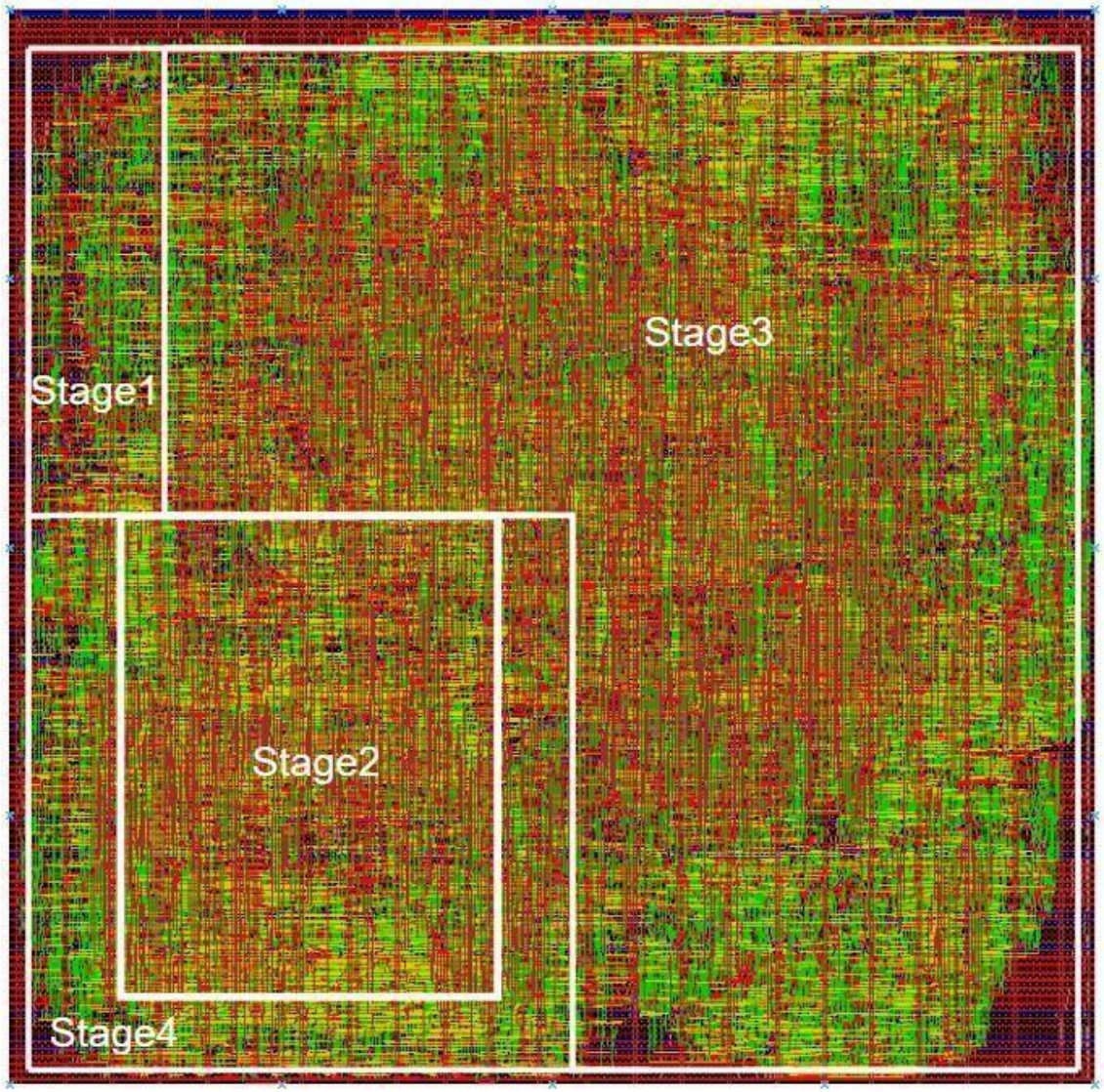
Highlight module(2/3)



Highlight module(3/3)

- 1.用ipad繪圖
- 2.用小畫家繪圖
- 3.用任意方法繪圖

- 大致繪出各級面積即可

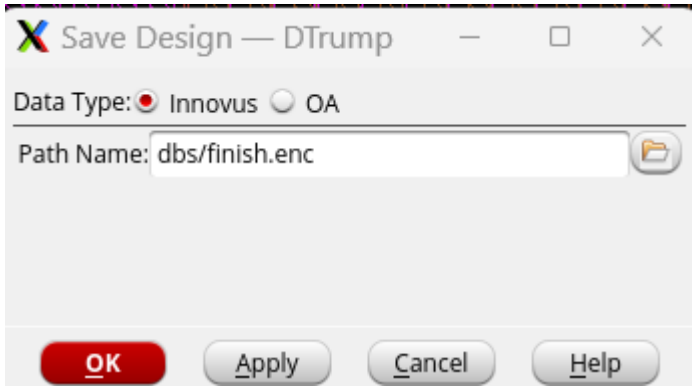


Finish

- **Save Design**

  File →Save Design

  dbs/finish.enc



Thank you